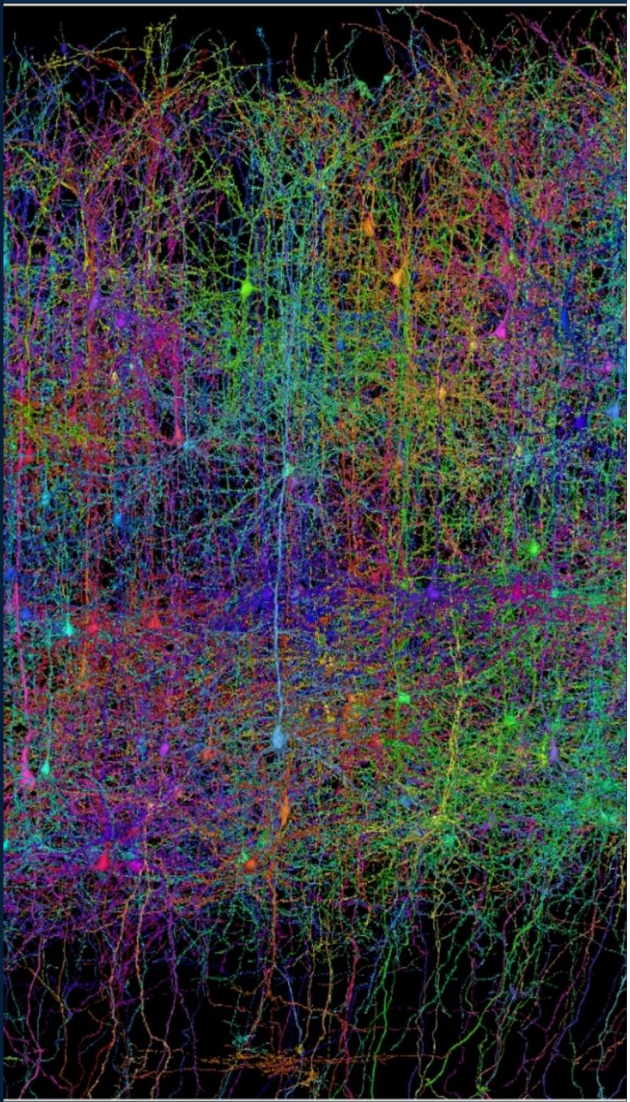
INTERNSHIP REPORT

Feature-Enhanced Statistical Modeling of Cortical Connectivity in Visual Cortex

Latent representations for functional connectomics

APRIL - AUGUST 2026 | 4 MONTHS

Milan, Italy



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Obviously, an internship in Milan would not have been quite the same if it had consisted only of adjacency matrices and a few million pairs of neurons. So thank you to everyone I met during these months who made the human side of this professional experience at least as memorable as the scientific one.

To the French running club, to David, Lola and everyone else: see you soon for that famous Monza–Bergamo night run with headtorches. Now that it is in black and white in a thesis, it becomes legally much more difficult to back out.

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And finally, to everyone with whom I shared a run, a meal, a drink, an evening, a match, an absurd conversation or simply a few hours of this Milanese interlude. It would be slightly dishonest to claim that you all contributed to the scientific quality of this thesis; on the other hand, you contributed enormously to making these few months a period I am unlikely to forget.

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1. Host institution and scientific context of the internship

1.1. Bocconi University and the computational neuroscience team

Bocconi University is primarily known for its programmes in economics, management and the social sciences. The presence of a group working on visual cortex may therefore seem rather surprising. It can be explained by the creation, in spring 2022, of the **Department of Computing Sciences**, which brings together researchers in computer science, applied mathematics, theoretical physics, artificial intelligence and machine learning. Creation of the Department of Computing Sciences [1]

Computational neuroscience is one of the research areas of this new department. This still relatively small team includes Carlo Baldassi, Nicolas Brunel and Alessandro Sanzeni. It seeks to understand how the brain processes sensory information, forms memories or makes decisions by combining mathematical models and tools from physics and machine learning with experimental data. Research areas - Department of Computing Sciences [2]

Alessandro Sanzeni, one of my internship co-supervisors, is particularly interested in how the brain processes visual information. When an animal looks at an image, the neurons in its visual cortex do not all respond in the same way: some respond more strongly to an orientation, a movement or a particular region of its visual field. These responses depend on the characteristics of the image, but also on the connections between neurons. His group therefore seeks to understand how the organisation of these connections influences the collective processing of information. To do so, it develops mathematical models linking the images presented, the activity recorded and the anatomy of the neuronal network. The objective is not merely to predict a neuron's response, but to identify the biological mechanisms that might explain it.

Several recent publications illustrate this direction. A study presented at the **International Conference on Learning Representations** (ICLR) in 2025 examines *digital twins* of visual cortex. The authors show that accurate predictions for each individual neuron do not guarantee a faithful reproduction of their collective activity. The evaluation of these models must therefore consider both individual responses and the way they are organised at network level. Liscai et al., ICLR 2025 [3]

Another paper, published in 2025 with Nicolas Brunel, investigates the presence of vestibular information in primary visual cortex. The authors show that V1 receives real-time information about the direction, speed and acceleration of head movements. Visual cortex therefore does not process only information coming from the eyes: its activity also integrates the context provided by the animal's movements. Bouvier et al., PNAS 2025 [4]

Finally, a study based directly on MICrONS data, a vast project devoted to the mouse visual cortex, examines the links between neurons' visual selectivity and their anatomical connectivity at synaptic resolution. It forms the scientific starting point of my internship, which approaches this relationship from a predictive perspective: determining which properties of two neurons make it possible to anticipate whether a connection exists between them. Buendía, Biggiogera and Sanzeni, 2025 [5]

The group also develops its own tools for working with this dataset. Victor Buendía, a member of the team, is the creator of MICrONS-combiner, a Python library designed to facilitate the downloading, organisation and processing of the various MICrONS tables. MICrONS-combiner - GitHub [6]

The library was still under active development during my internship. As one of its first users, I was therefore able to take part in its beta testing by applying it to a real use case. This work led me to identify and document unexpected behaviour, then pass this feedback on to the team in order to help improve the tool.

The project was also co-supervised by Alberto Antonietti, a lecturer and researcher at Politecnico di Milano and a member of **NEARLab**. A biomedical engineer by training, he builds computational models from experimental data that are intended to reproduce the electrical behaviour of neurons and the networks they form. His expertise therefore made it possible to place my work on connectivity in the broader context of neuronal-circuit simulation. Alberto Antonietti - Politecnico di Milano [7]

1.2. Organisation and supervision of the internship

This scientific environment defined the general framework of my internship. Because the project was largely exploratory, we still needed to establish a suitable working method. Drawing on lessons from my previous experiences, even before arriving I had expressed two needs: access to a workspace with considerable flexibility in working hours, and regular supervision.

We therefore agreed with Professor Sanzeni to organise a weekly meeting. In return for the autonomy I was given, I undertook to prepare a clear and detailed account of my progress each week, generally in the form of a PowerPoint presentation or a PDF.

Every Monday afternoon, I met with Professors Sanzeni and Antonietti for approximately one hour. I presented the results obtained during the week, the difficulties encountered and the questions that remained open. These discussions allowed us to consider their reservations, rule out certain hypotheses and define new directions for the work. These successive reports gradually came to form a genuine logbook of the internship.

A desk had also been made available to me at Bocconi, together with a Discord channel on which I could flag a difficulty more quickly or share an intermediate result. My supervisors were always particularly available and responsive. I therefore did not follow strictly defined working hours: my work was assessed mainly through my progress and my ability to present it clearly at the following meeting. This balance between organisational freedom and an obligation to deliver results suited the way I work.

This autonomy did not, however, mean that I was working alone. Although machine learning was not unfamiliar to me, neuroscience was almost entirely new. At the beginning of the internship, reading papers was challenging because of the highly specialised vocabulary. Concepts such as the connectome, *proofreading* and the various neuronal structures still required a real effort of understanding on my part. My learning then drew on three complementary sources: scientific publications, laboratory meetings and the courses I was able to attend.

Every Tuesday, during the lunch break, I was invited to the *NeuroML* meetings. A researcher from the department or a guest speaker would present their work, a method or a recent paper. The first presentations were sometimes difficult to follow, but they gradually became more accessible as I acquired the necessary vocabulary. I almost regret that I only began to appreciate their full value towards the end of the internship.

Professor Antonietti also invited me to Politecnico di Milano on Friday mornings to attend thesis and project presentations over a pastry. These were generally more oriented towards neuroengineering and systems modelling, with, for example, work on the control of a bionic arm. They offered a more applied perspective than the mainly theoretical research presented at Bocconi.

Finally, I attended several courses and presentations organised at Bocconi, including a session by Victor Buendía on the use of MICrONS-combiner. These different forms of learning gradually helped me understand the field, but also to place my own results within a broader scientific context.

This framework, combining autonomy, weekly supervision and continuous learning, structured my entire internship. It proved particularly important during the first few weeks, which were devoted to discovering the MICrONS data and reproducing the work that would serve as the starting point for my project.

1.3. The MICrONS project and functional connectomics

“A cubic millimetre might not sound like much — roughly the size of a grain of sand — but to neuroscientists, it is enormous.”

This is how *Nature* introduced the MICrONS project in an article published in April 2025. One cubic millimetre may indeed sound ridiculously small: it is approximately the size of a grain of sand. Yet, at the scale of the brain, this tiny volume already contains tens of thousands of neurons linked by hundreds of millions of synapses. Reconstructing all these cells and their connections is therefore anything but a small task. The MICrONS project - Nature [8]

MICrONS stands for *Machine Intelligence from Cortical Networks*. The project grew out of a particularly ambitious idea: to observe how a real cortical network is organised and how it responds to different stimuli, with enough precision to extract principles that might inspire new machine-learning algorithms.

The project focused on approximately one cubic millimetre of a mouse's visual cortex. Although this volume represents only a tiny fraction of the brain, the resulting dataset contains approximately **200,000 cells**, **523 million synapses**, together with functional recordings from around **75,000 neurons**. [9]

What makes MICrONS original is not only its scale, but also its combination of two complementary observations of the same cortical tissue. On the one hand, electron microscopy provides an extremely detailed anatomical reconstruction of the neurons, their axons, dendrites and synapses. On the other, two-photon calcium imaging records the response of some of these neurons while the mouse views different visual stimuli.

MICrONS therefore provides more than a static map of brain wiring. For some of the neurons, it becomes possible to ask two questions simultaneously: **How is this neuron connected? How does it respond to visual information?**

Bringing these two questions together is the central point of functional connectomics. Instead of studying neuronal activity and anatomical connectivity separately in different experiments, MICrONS makes it possible to relate them at the level of individual neurons and their synapses.

neurons and synapses.

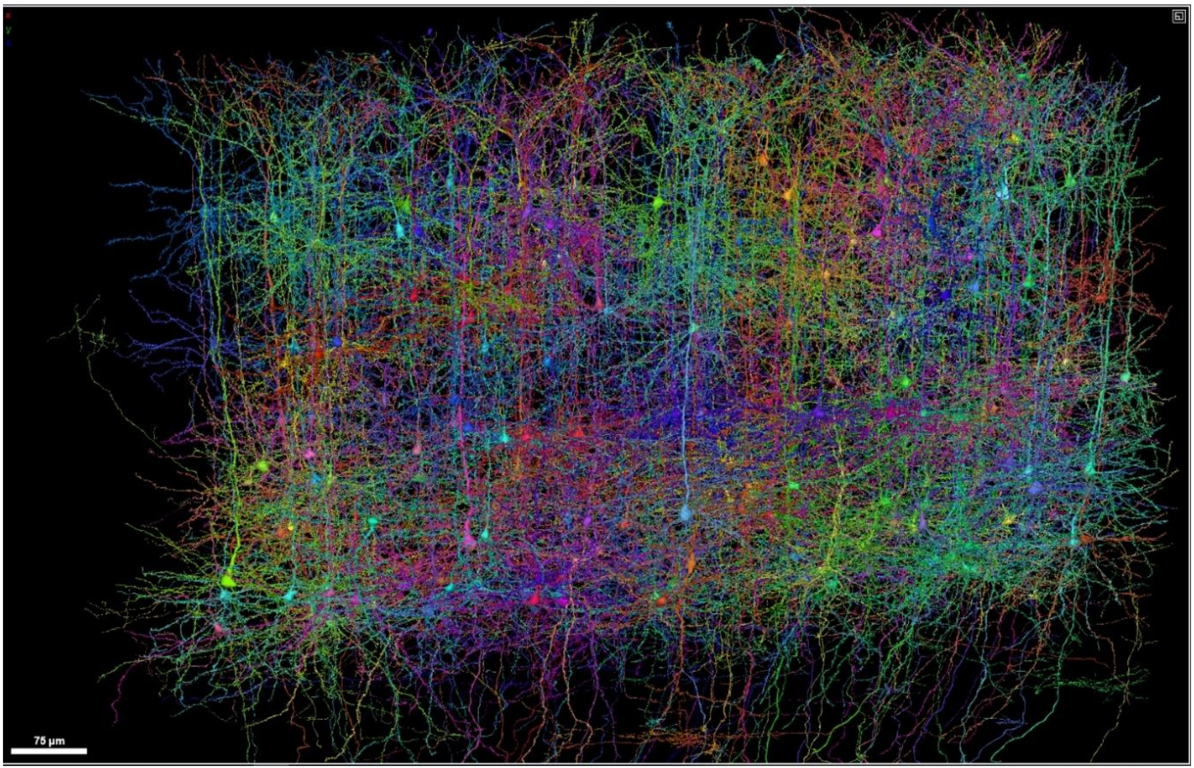


Figure 1 - Three-dimensional reconstruction of neurons in the visual-cortex volume studied by MICrONS.

1.3.1. Recording neuronal activity

The genetically modified mouse used in the experiment expressed the GCaMP6s protein in its excitatory neurons. This protein is only weakly fluorescent at rest, but becomes much brighter when it binds calcium. Since calcium influx generally accompanies neuronal activity, changes in fluorescence provide an indirect measure of this activity.

During the experiment, the mouse was awake and its head was held beneath the microscope, but it could run on a cylindrical treadmill. A screen positioned in front of its left eye displayed different visual stimuli: film clips, sports videos, virtual environments and synthetic patterns designed to measure neurons' sensitivity to orientation and direction of motion.

Activity was recorded using two-photon calcium imaging. An infrared laser beam successively scanned several planes of the cortex. The value of this technique lies in focusing the beam within an extremely small volume. To be excited, a GCaMP6s molecule must absorb two photons almost simultaneously. Yet this double absorption is highly unlikely when photon density is low.

Thus, when a photon encounters a calcium-bound GCaMP6s molecule before reaching the focal point, it is generally not sufficient to excite it. It is primarily around the focal point that the instantaneous photon density becomes high enough for two-photon absorption to occur. Fluorescence therefore comes mainly from a small, well-localised volume, making it possible to observe neurons located as deep as approximately $500\ \mu\mathrm{m}$ below the cortical surface while limiting fluorescence from the tissue traversed.

The microscope did not therefore target neurons manually one by one. It scanned complete planes, from which the cell bodies were then segmented computationally. For each neuron detected, a fluorescence trace was extracted over time. Neuronal activity was quantified through the relative change in fluorescence.

1.3.2. Reconstructing the anatomical architecture

Once the functional recordings were complete, the mouse was euthanised and its brain fixed by transcardial perfusion. The region previously observed using calcium imaging was then relocated using the blood vessels visible at the surface of the cortex and extracted. It was treated with heavy metals in order to increase membrane contrast under an electron microscope.

The tissue was dehydrated and then infiltrated with epoxy resin. As it hardens, this resin makes it possible to cut the sample without distorting its organisation. An ultramicrotome thus divided the block into 27,972 consecutive sections approximately $40\ \mathrm{nm}$ thick.

Of these sections, 26,652 were imaged over approximately six months using five automated transmission electron microscopes, known as *autoTEMs*. An electron beam passes through each ultrathin section. Regions containing more heavy metals scatter the electrons more strongly and appear darker, making it possible to distinguish the membranes of somas, axons and dendrites, as well as the characteristic structures of synapses. The resolution obtained was close to $4\ \mathrm{nm}$ per pixel in the plane of each image.

The thousands of two-dimensional images were then corrected, aligned and stacked in the order in which they had been cut in order to reconstruct the cortical volume digitally in three dimensions.

Machine-learning algorithms finally segmented the cells, detected synapses and identified their presynaptic and postsynaptic partners. This automatic reconstruction could nevertheless merge two different neurons or artificially interrupt a neuronal process. Some cells were therefore checked and corrected manually or semi-automatically in a step known as *proofreading*.

I was rather surprised by the apparent simplicity of the principles used for these two acquisitions. I had initially expected functional MRI to measure activity and some form of tomographic scanner to reconstruct the anatomy, as in clinical imaging.

2. State of the art and reproduction of the reference model

When I arrived at Bocconi University in April, Professor Sanzeni was supervising the master's thesis of Alessandro Butti, a student at Politecnico di Milano. This work, then almost complete, was entitled *Emergence of Spatially Structured Orientation Tuning in Connectomics-Based Spiking Network Models of V1*. [10]

To familiarise myself with the MICrONS connectome while assessing the reproducibility of this work, he suggested that I try to reproduce its main results independently. I did not have access to the code developed by Alessandro: I had to rebuild the analysis pipeline myself from the dataset, his thesis and the results presented in it.

This exercise therefore provided a good introduction. It allowed me both to familiarise myself with the complex structure of the MICrONS data, to understand the methodological choices made in the previous work and to check whether its results could be recovered using an entirely new implementation.

2.1. Reference geometric model

Alessandro's thesis investigated the relationship between the structure of visual cortex and neurons' orientation selectivity. Some neurons do indeed respond more strongly to a particular orientation. A classical hypothesis, known as *like-to-like connectivity*, assumes that neurons sharing similar preferences are more likely to be connected to one another. [11]

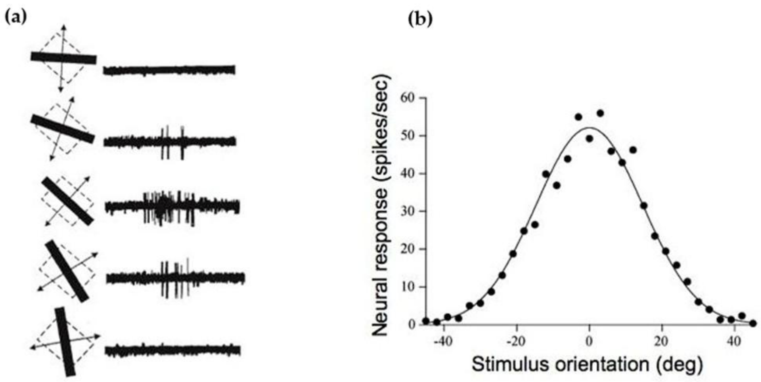


Figure 2 - Orientation selectivity in primary visual cortex. (a) Response of a V1 neuron to bars presented at different orientations. Its firing rate is highest for a particular orientation, known as its preferred orientation. (b) Corresponding tuning curve, showing the neuron's response as a function of stimulus orientation. Adapted from Hubel and Wiesel (1959) [12].

To test this hypothesis, Alessandro drew on layers 2/3 and 4 of the MICrONS connectome. Since the latter contains reconstruction errors and had only been manually corrected for a small proportion of the neurons, it could not be used directly as a perfectly reliable network. From the *proofread* neurons, he therefore learned statistical rules linking neuronal position to connection probability and estimated synapse size. These rules were then used to generate synthetic networks, whose activity was simulated under different visual stimuli.

His main result is that adding the difference in orientation improves connection prediction very little compared with geometry alone. In the volume studied, nearby neurons are both more likely to be connected and often sensitive to similar orientations. A rule based on spatial proximity therefore already produces part of the *like-to-like* effect, without orientation having to be used directly to establish connections.

When this wiring is incorporated into a spiking neural network and combined with orientation-selective inputs from L4, the simulation produces a spatial organisation of orientation preferences in L2/3 similar to that observed in MICrONS. The work thus suggests that circuit geometry, inputs from L4 and recurrent dynamics can together reproduce part of the functional organisation of V1, without having to impose a connection rule based directly on orientation similarity.

2.1.1. Biological organisation of visual cortex

Before beginning the analysis, I first had to understand how the connectome and, more generally, the cerebral cortex were organised. The superficial portion of the cortex is composed of grey matter, rich in neuronal cell bodies. It is divided into six layers, numbered from L1, the most superficial, to L6, the deepest. Beneath the cortex lies the white matter, composed mainly of myelinated axons that transmit information between different regions of the brain.

In our case, we were interested in area V1, the first cortical area involved in processing visual information, and more specifically in layers L4 and L2/3. Information captured by the retina first passes through the thalamus before primarily reaching layer L4 of V1. This layer then transmits the information to layers L2/3, which continue to process it and pass it on, in particular, to other cortical regions.

Each layer contains excitatory and inhibitory neurons. The former generally increase the probability that the neurons to which they are connected will produce an action potential, while the latter reduce that probability. The balance between excitation and inhibition is essential to network stability: insufficiently controlled excitation can lead to excessive and highly synchronised activity, such as that observed during certain epileptic seizures.

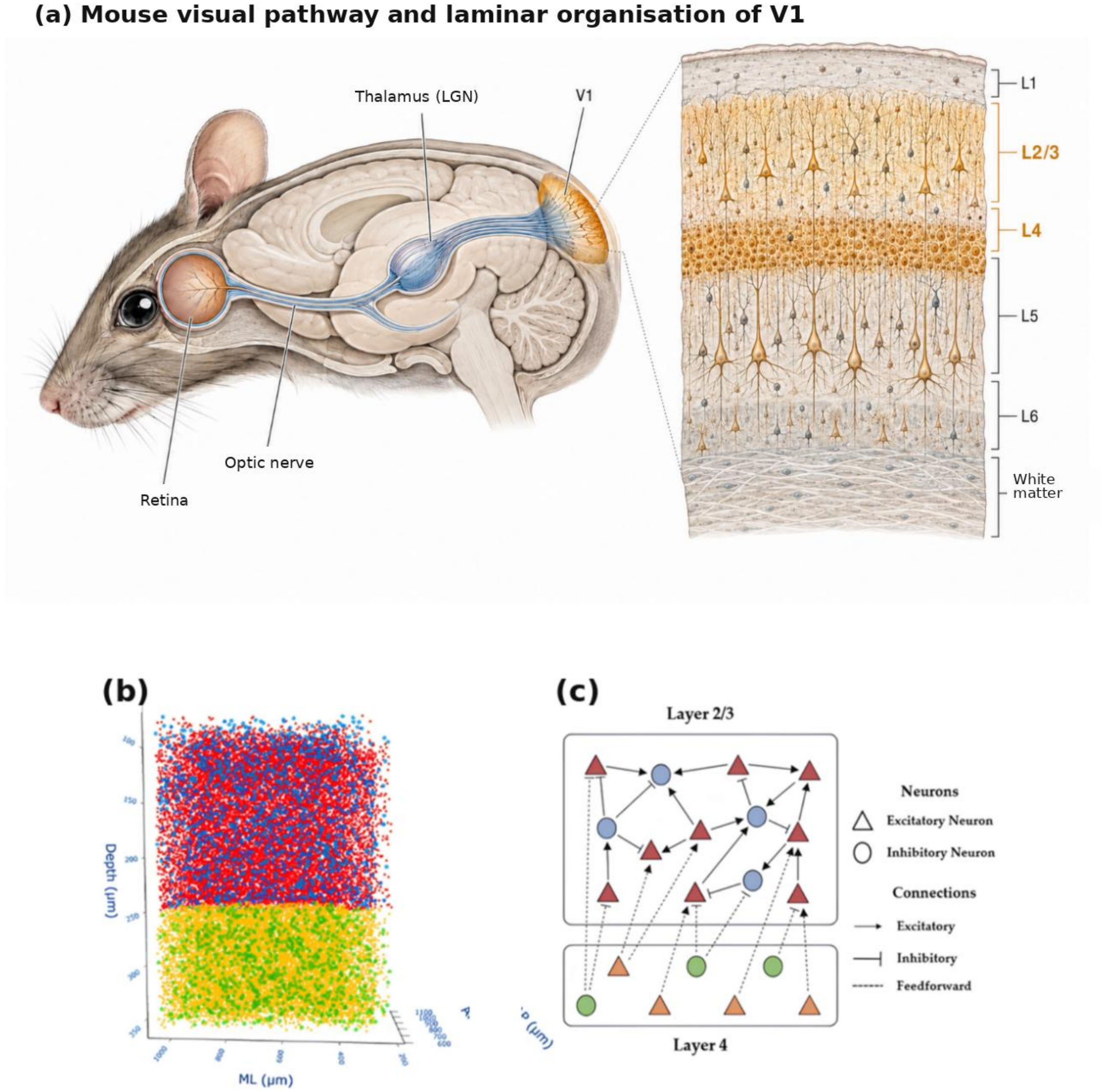


Figure 3 - From the visual pathway to the cortical subcircuit studied. (a) Simplified organisation of the visual pathway and area V1 in the mouse. Information captured by the retina passes through the thalamus before reaching V1, whose L2/3 and L4 layers are highlighted. (b) Three-dimensional distribution of excitatory and inhibitory neurons in L2/3 and L4 within the volume studied. (c) Simplified representation of the subcircuit considered, including recurrent excitatory and inhibitory connections within L2/3 and feedforward projections from L4 to L2/3.

2.2. Dataset construction

Running this code seems relatively simple in hindsight, but it had appeared considerably more complex to me at the time. Since this first processing pipeline will be useful later in the thesis, I think it is worth describing it here. We begin by loading the units_L234 table, which brings together anatomical information and some functional information from the two experiments described above. Each neuron is listed with various characteristics, such as its size, position, layer, type and, when available, preferred orientation.

We then select the neurons belonging to layers L2/3 and L4, before separating them according to type, excitatory or inhibitory. Among the different possible configurations, we exclude only those in which an L4 neuron is postsynaptic, since we are studying recurrent connections within L2/3 and projections from L4 to L2/3, which corresponds to one of the main pathways through which visual information propagates within V1. We extract the information of interest to us, notably the neurons' position and preferred orientation.

(a) Selection by type and layer

```
mask_exc = (units_cleaned['classification_system'] == 'excitatory_neuron')
mask_inh = (units_cleaned['classification_system'] == 'inhibitory_neuron')

mask_l23 = (units_cleaned['layer'] == 'L2/3')
mask_l4 = (units_cleaned['layer'] == 'L4')
```

(b) Creation of the four populations

```
E_L23 = units_cleaned[mask_exc & mask_l23].copy().reset_index(drop=True)
I_L23 = units_cleaned[mask_inh & mask_l23].copy().reset_index(drop=True)
E_L4 = units_cleaned[mask_exc & mask_l4].copy().reset_index(drop=True)
I_L4 = units_cleaned[mask_inh & mask_l4].copy().reset_index(drop=True)
```

(c) Definition of the eight pathways

```
pathways = [
    # post = E_L23
    ('E23_E23', E_L23, E_L23),
    ('I23_E23', I_L23, E_L23),
    ('E4_E23', E_L4, E_L23),
    ('I4_E23', I_L4, E_L23),

    # post = I_L23
    ('E23_I23', E_L23, I_L23),
    ('I23_I23', I_L23, I_L23),
    ('E4_I23', E_L4, I_L23),
    ('I4_I23', I_L4, I_L23),
]
```

Figure 4 - Definition of the neuronal populations and connection pathways studied. (a) Construction of masks to separate neurons according to their type, excitatory or inhibitory, and their layer, L2/3 or L4. (b) Application of these masks to the units_cleaned table to obtain the four populations $E_{L2/3}$, $I_{L2/3}$, E_{L4} and I_{L4}, where E and I respectively denote excitatory and inhibitory neurons. (c) Definition of the eight pathways studied. Each triplet contains the name of the pathway, the presynaptic population and the postsynaptic population. The configurations retained correspond to recurrent connections within L2/3 and feedforward projections from L4 to L2/3.

We then use the synapses_L234 table. This is a relatively simple table containing three main pieces of information for each synapse: the identifier of the presynaptic neuron, the identifier of the postsynaptic neuron and the estimated synapse volume. The presence of at least one synapse between two neurons means that their connection can be considered observed. We then retain only connections for which both the presynaptic and postsynaptic neurons belong to the previously selected populations.

Denoting by $\mathcal{V}pre$ the set of presynaptic neurons, by $\mathcal{V}$post the set of postsynaptic neurons and by S the set of observed connections, the set of negative examples can be written as:

$$S_{neg} = (\mathcal{V}\mathrm{pre} \times \mathcal{V}\mathrm{post}) \setminus S_{pos}$$

We also take care to remove autapses $(i = j)$, that is self-connections. Biologically, they do exist but including them in the model would make training far more complex, in particular because their intersomatic distance is zero.

This exhaustive construction has at least quadratic complexity with respect to the number of neurons.

$$\mathrm{C} = \Theta(n^2)$$

This growth quickly becomes restrictive: the connectome contains relatively few observed connections, but several million potential pairs. Negative examples therefore represent the vast majority of the dataset.

For each pair of neurons, connected or not, we calculate two quantities: the spatial distance between their somas and the difference between their preferred orientations. The first is simply the Euclidean distance between their positions in the cortex. The second requires greater care, because an orientation is defined modulo π.

To calculate this difference in orientation, Alessandro used the following formula:

$$\Delta\theta(a,b) = \left(\left(a - b + \frac{\pi}{2}\right) \bmod \pi\right) - \frac{\pi}{2}$$

The idea is to transform the difference between two orientations into a value that the model can interpret correctly. For example, if $a = \pi/2$ and $b = -\pi/2$, ordinary subtraction produces a difference of π. Yet these two angles represent exactly the same orientation, and their difference must therefore be zero.

I spent some time looking for a profound geometric explanation of this equation, but ultimately it is above all a clever folding operation. We begin with a difference (a-b) between - π and π. Since an orientation repeats every π radians, we take modulo π. We nevertheless want the discontinuity to occur between $-\pi/2$ and $\pi/2$ rather than between 0 and π. We therefore add $\pi/2$ before taking the modulo, then subtract it afterwards.

The transformation preserves the difference when it belongs to the interval $[-\pi/2, \pi/2[$. It adds π to values below $-\pi/2$ and subtracts π from values greater than or equal to $\pi/2$. All differences are thus folded into the same interval:

$$\Delta\theta(a,b) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right[$$

Since I had initially found Alessandro's formulation difficult to understand - it does give you a bit of a headache when you first come across it - I had started by using an approach that seemed more geometric to me.

I represented each orientation by a point on a circle of radius 1, then measured the Euclidean distance between the two points. More precisely, an orientation θ was represented by:

$$(\cos(2\theta), \sin(2\theta))$$

Doubling the angles ensures that $\theta$and $\theta + \pi$ correspond to the same point on the circle and therefore represent the same orientation.

This method does indeed make it possible to measure the similarity between two orientations, but it returns a chord length between 0 and 2 rather than a signed angular difference expressed in radians. It therefore loses the information about the sign of (a-b) and is not exactly the same variable as the one used by Alessandro, although it remains suitable for quantifying the proximity between two orientations.

We then prepare the dataset by assigning the label (y=1) to all pairs for which a connection is observed in the connectome, and (y=0) to those for which no connection is recorded. The second label therefore represents the absence of an observed connection, and not necessarily a genuine biological absence, given the possibility of reconstruction errors.

2.3. Model, training and loss function

We then create the model that will allow us to estimate connection probability. It is a *Multi-Layer Perceptron* (MLP), one of the simplest artificial neural-network architectures.

Given the univariate input, either Euclidean distance or angular difference, the architecture selected is deliberately minimal:

Architecture of the univariate MLP

```
class MLP_v2(nn.Module):
    def __init__(self):
        super().__init__()

        self.net = nn.Sequential(
            nn.Linear(1, 8),
            nn.ReLU(),
            nn.Linear(8, 4),
            nn.ReLU(),
            nn.Linear(4, 1),
            nn.Sigmoid()
        )

    def forward(self, x):
        return self.net(x)
```

Figure 5 - Architecture of the univariate MLP

The two hidden layers contain eight and four neurons respectively. The ReLU activations introduce the non-linearity required to learn a relationship between distance and connectivity that is more complex than a simple straight line. Without them, the succession of three linear layers would remain mathematically equivalent to a single linear transformation.

The sigmoid function then maps the model's output to the interval [0,1].

The MLP is technically a probabilistic classifier, but here it is used as an estimator rather than as a classifier producing a definitive decision. No threshold is applied to its output: the model does not directly answer "connected" or "not connected", but provides a connection probability for each pair of neurons.

To train this model, we use *binary cross-entropy* as the loss function. This choice becomes fairly natural if each potential connection is viewed as a Bernoulli trial.

For each ordered pair of neurons, we define a random variable Y_{ij} indicating whether a connection from one neuron to the other is observed:

$$Y_{ij} = \begin{cases} 1 & \text{if a connection from } i \text{ to } j \text{ is observed,} \\ 0 & \text{otherwise.} \end{cases}$$

This variable follows a Bernoulli distribution with parameter p_{ij}:

$$Y_{ij} \sim \text{Bernoulli}(p_{ij})$$

By definition of the Bernoulli distribution:

$$\begin{cases} \mathbb{P}\left(Y_{ij} = 1\right) = p_{ij}, \\ \mathbb{P}\left(Y_{ij} = 0\right) = 1 - p_{ij}. \end{cases}$$

These two possibilities can be combined in a single expression:

$$\mathbb{P}\left(Y_{ij} = y_{ij}\right) = p_{ij}^{y_{ij}}\left(1 - p_{ij}\right)^{1-y_{ij}}$$

Our objective is precisely to estimate this unknown parameter. We denote the probability estimated by the MLP as:

$$\tilde{p}_{ij} = \mathrm{MLP}_{\theta}\left(x_{ij}\right)$$

The set of probabilities estimated for the different pairs is then denoted as:

$$\tilde{\mathbf{p}} = \left\{\tilde{p}_{ij}\right\}_{(i,j)\in\mathcal{P}}$$

The observed adjacency matrix corresponds to the simultaneous realisation of all these trials. The event consisting in observing this exact matrix can therefore be written as the intersection of the events associated with each pair:

$$\{\mathbf{Y} = \mathbf{y}\} = \bigcap_{(i,j)\in\mathcal{P}} \left\{Y_{ij} = y_{ij}\right\}$$

Let us then consider the likelihood, that is, the probability of obtaining the observed adjacency matrix for a given set of Bernoulli parameters:

$$\mathcal{L}(\tilde{\mathbf{p}}) = \mathbb{P}(\mathbf{Y} = \mathbf{y} \mid \tilde{\mathbf{p}})$$

Intuitively, it is easy to understand that this function is maximal when the estimated probabilities $\tilde{\mathbf{p}}$ correspond to the true probabilities $\mathbf{p}$. We therefore seek the set of probabilities that maximises the likelihood:

$$\tilde{\mathbf{p}}^{\star} = \arg\max_{\tilde{\mathbf{p}}} \mathcal{L}(\tilde{\mathbf{p}})$$

Assuming that the different Bernoulli trials are mutually independent, the likelihood decomposes into a product of individual probabilities:

$$\mathcal{L}(\tilde{\mathbf{p}}) = \mathbb{P}(\mathbf{Y} = \mathbf{y} \mid \tilde{\mathbf{p}}) = \prod_{(i,j)\in\mathcal{P}} \mathbb{P}\left(Y_{ij} = y_{ij} \mid \tilde{p}_{ij}\right)$$

Using the preceding expression for the Bernoulli distribution, we obtain:

$$\mathcal{L}(\tilde{\mathbf{p}}) = \prod_{(i,j)\in\mathcal{P}} \tilde{p}_{ij}^{y_{ij}}\left(1 - \tilde{p}_{ij}\right)^{1-y_{ij}}$$

Since this is a product, we take its logarithm in order to turn it into a sum. As the logarithm is strictly increasing, this operation does not change the location of the maximum. Finally, since optimisation algorithms are generally formulated as minimisation problems, we minimise the negative log-likelihood.

We thereby obtain binary cross-entropy:

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{N}\log \mathcal{L}\left(\tilde{\mathbf{p}}\right) = -\frac{1}{N}\sum_{(i,j)\in\mathcal{P}}\left[y_{ij}\log\left(\tilde{p}_{ij}\right) + \left(1-y_{ij}\right)\log\left(1-\tilde{p}_{ij}\right)\right]$$

Since each probability depends directly on the MLP parameters, minimising this function ultimately amounts to optimising those parameters:

$$\theta^{\star} = \arg\min_{\theta}\ \mathcal{L}_{\mathrm{BCE}}(\theta)$$

We then optimise the MLP parameters using Adam, a more sophisticated version of gradient descent that nevertheless retains the same principle.

At each iteration, we calculate the gradient of the BCE with respect to the model parameters, then update those parameters according to:

$$\theta_{i+1} = \theta_i - \eta\nabla_{\theta}\mathcal{L}_{\mathrm{BCE}}(\theta_i), \qquad \eta = 10^{-3}$$

The gradient indicates the direction in which the loss function increases most rapidly. We therefore move in the opposite direction.

The learning rate η controls the magnitude of each update: a value that is too high could make training unstable, while a value that is too low would slow convergence.

Training loop

```
for pi, pj, yb in train_loader:
    p_observed = model(pi, pj)
    loss = criterion(p_observed, yb)

    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

    total_train_loss += loss.item() * len(yb)
```

Figure 6 - MLP training loop and parameter update.

In practical terms, zero_grad() clears the gradients accumulated during the previous batch, backward() calculates the loss gradients for the current batch, then step() uses these gradients to update the parameters according to the rule defined by Adam.

2.4. Evaluation protocol

Now that we have a functional model theoretically capable of learning connection probabilities, we need a metric with which to assess the accuracy of that learning.

The first evaluation is mainly qualitative, although it could easily be made quantitative. It consists in checking whether the model reproduces a relationship between connection probability and distance similar to the one observed empirically.

To do so, we place all pairs of neurons, connected or not, into different distance bins. For each bin, the empirical probability is the ratio of the number of positive pairs, that is, pairs with an observed connection, to the total number of pairs:

$$\hat{p}_k^{\mathrm{emp}} = \frac{\text{number of positive pairs in bin } k}{\text{total number of pairs in bin } k}$$

This gives us our empirical reference curve.

To construct the curve predicted by the model, we calculate, in each bin, the mean of the connection probabilities $\tilde{p}_{ij}$ estimated by the MLP:

$$\hat{p}_k^{\mathrm{mod}} = \frac{\sum_{(i,j)\in\mathcal{B}_k} \tilde{p}_{ij}}{\text{total number of pairs in bin } k}$$

If the two curves are close, it means that the model has correctly learned the overall relationship between distance and connectivity. This comparison does not yet tell us, however, whether at the level of each pair it correctly distinguishes an observed connection from a non-connection. This second question will be studied using the AUC.

The AUC is widely used to evaluate the performance of a binary classifier. However, since my supervisor was not entirely familiar with this tool, I had prepared a short slide deck to explain how it works. I therefore think that a brief refresher for the reader is not superfluous.

To evaluate the model as a classifier, we turn the estimated probabilities into decisions using a threshold τ: a pair is predicted to be connected if $\tilde{p}_{ij} \geq \tau$.

Although there is an infinite number of thresholds between 0 and 1, the decisions change only when τ crosses one of the predicted probabilities. It is therefore sufficient to consider the distinct values

$\tau_k = \tilde{p}_{\mathrm{pair\ k}}$

For each threshold, we construct a confusion matrix, that is, a table crossing the true state of each pair with the model's decision:

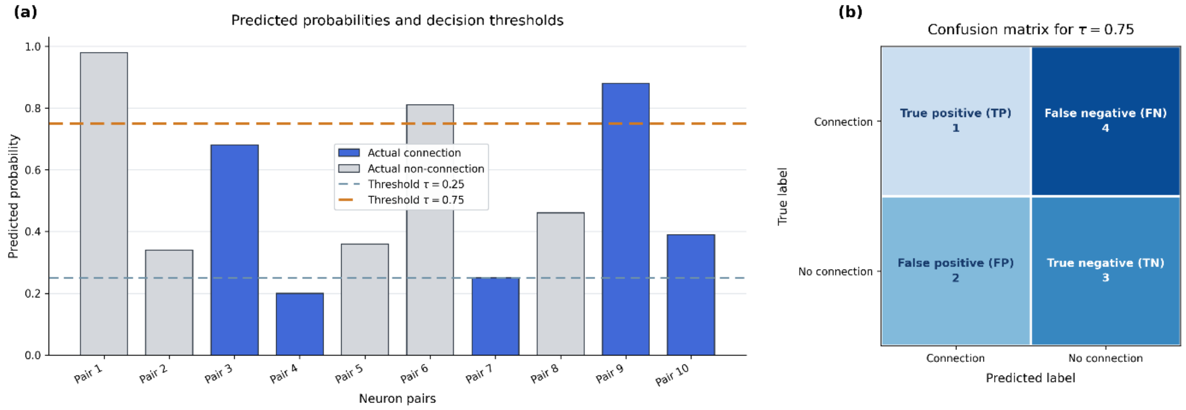


Figure 7 - Construction of a confusion matrix from the predicted probabilities. Panel (a) shows the probabilities assigned to the different pairs of neurons together with two decision thresholds, while panel (b) shows the confusion matrix obtained for $\tau = 0{,}75$.

From this we derive the proportion of connections correctly detected (True Positive Rate):

$$\mathrm{TPR}(\tau_k) = \frac{\mathrm{TP}(\tau_k)}{\mathrm{TP}(\tau_k) + \mathrm{FN}(\tau_k)}$$

And the proportion of non-connections incorrectly predicted as connected (False Positive Rate):

$$\mathrm{FPR}(\tau_k) = \frac{\mathrm{FP}(\tau_k)}{\mathrm{FP}(\tau_k) + \mathrm{TN}(\tau_k)}$$

Each threshold then provides one point on the ROC curve, with the FPR on the x-axis and the TPR on the y-axis:

$$R_k = (\mathrm{FPR}(\tau_k)\,, \mathrm{TPR}(\tau_k))$$

This construction can be understood more intuitively by starting from $\tau = 0$. All pairs are then predicted as connected: the TPR and FPR therefore both have the same value, and we start from the point $R_0 = (1{,}1)$.

When the threshold increases and crosses the score of a real connection, the latter changes from a *True Positive* to a *False Negative*. The TPR therefore decreases by $\frac{1}{N_+}$, producing a downward move. When it crosses the score of a non-connection, the latter changes from a *False Positive* to a *True Negative*. The FPR decreases by $\frac{1}{N_-}$, producing a move to the left.

A random classifier is equally likely to rank a pair correctly or incorrectly, making downward and leftward moves equally likely. Its ROC curve therefore follows the diagonal.

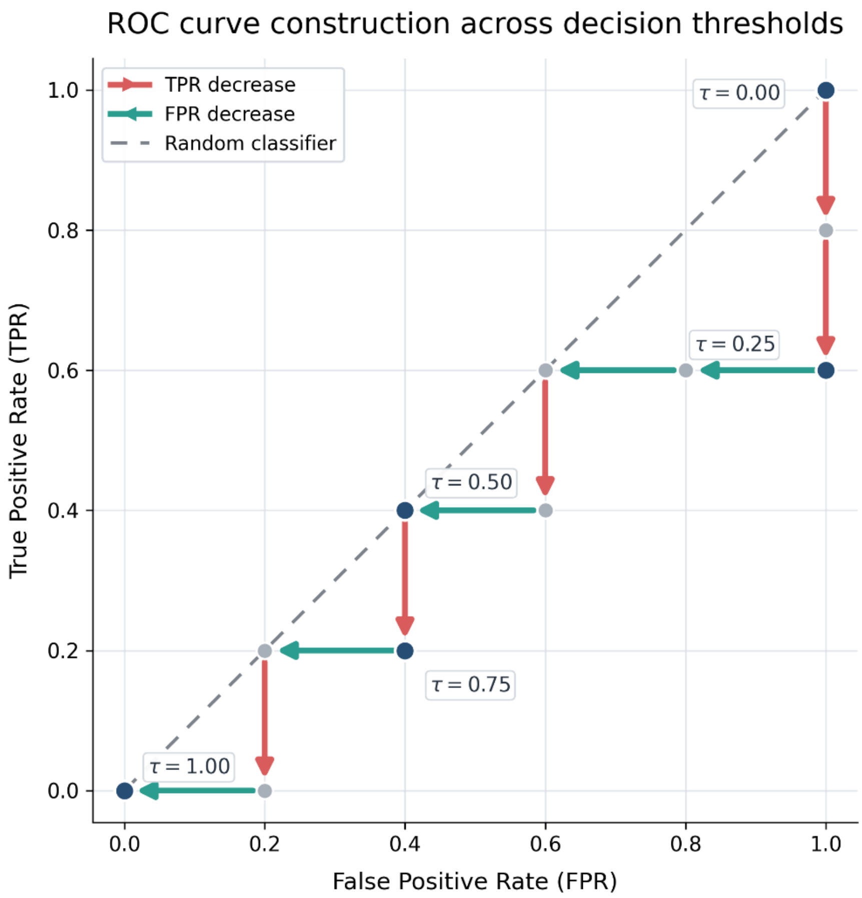


Figure 8 - Construction of the ROC curve as the decision threshold increases from 0 to 1. Crossing the score of a true connection decreases the TPR, represented by a red vertical move, while crossing that of a non-connection decreases the FPR, represented by a green horizontal move; the diagonal corresponds to the performance of a random classifier.

The AUC (*Area Under the Curve*) is the area beneath the ROC curve. It therefore aggregates the performance obtained across all possible thresholds, rather than evaluating the model for a particular threshold:

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(u)\ \mathrm{d}u, \qquad u = \mathrm{FPR}$$

It also has a probabilistic interpretation: if a real connection and a non-connection are selected at random, the AUC is the probability that the model assigns the higher score to the real connection, counting ties as half:

$$\mathrm{AUC} = \mathbb{P}(\tilde{p}^+ > \tilde{p}^-) + \frac{1}{2}\mathbb{P}(\tilde{p}^+ = \tilde{p}^-)$$

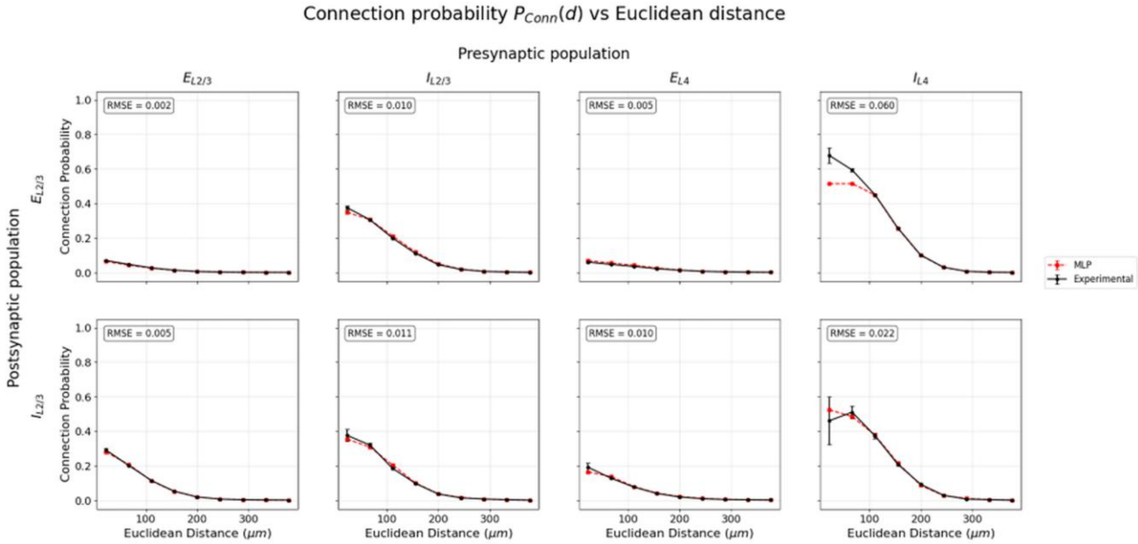
An AUC of 1 therefore corresponds to a perfect ranking. An AUC of 0.5 describes a random ranking, whose ROC curve follows the diagonal. The AUC thus measures the model's discriminative ability.

2.5. Reproduction results

Initially, I restricted myself to the model using only intersomatic distance. On the very first attempt, I obtained results very close to Alessandro's: our connection-probability curves had similar profiles and the AUCs differed by only a few hundredths.

I therefore did not consider it necessary to continue hyperparameter optimisation with Optuna. The purpose of this first step was not to obtain the best-performing model possible, but to verify the reproducibility of its results and familiarise myself with processing the connectome.

(a) Present work



(b) Alessandro's original results

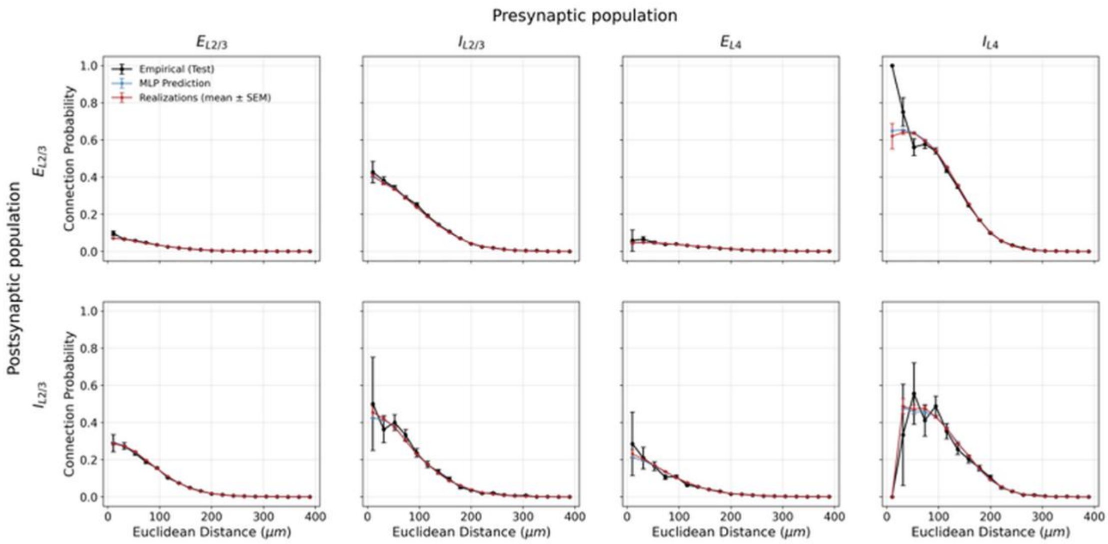


Figure 9 - Comparison of empirical and predicted connection probabilities as a function of Euclidean distance for the eight pre- and postsynaptic configurations: (a) results reproduced in this work; (b) Alessandro's original results.

(a) Present work

Pathway	Test BCE	Test AUC	# Test pairs
$E_{L2/3} \to E_{L2/3}$	0.0366	0.8474	1,105,104
$I_{L2/3} \to E_{L2/3}$	0.1607	0.8623	86,837
$E_{L4} \to E_{L2/3}$	0.0375	0.7981	1,286,755
$I_{L4} \to E_{L2/3}$	0.1420	0.9129	78,942
$E_{L2/3} \to I_{L2/3}$	0.0949	0.8827	109,116
$I_{L2/3} \to I_{L2/3}$	0.1470	0.8652	8,567
$E_{L4} \to I_{L2/3}$	0.0483	0.8454	127,043
$I_{L4} \to I_{L2/3}$	0.1242	0.9027	7,794

(b) Alessandro's original results

Pathway	Test BCE	Test AUC	# Test pairs
$E_{L2/3} \to E_{L2/3}$	0.0466	0.8219	611,209
$I_{L2/3} \to E_{L2/3}$	0.1508	0.8871	130,255
$E_{L4} \to E_{L2/3}$	0.0448	0.7752	607,371
$I_{L4} \to E_{L2/3}$	0.1430	0.9107	118,413
$E_{L2/3} \to I_{L2/3}$	0.0915	0.8944	163,674
$I_{L2/3} \to I_{L2/3}$	0.1409	0.8820	12,851
$E_{L4} \to I_{L2/3}$	0.0477	0.8504	190,564
$I_{L4} \to I_{L2/3}$	0.1226	0.9055	11,691

Figure 10 - Comparison of the performance obtained on the eight connectivity pathways: (a) results reproduced in this work; (b) Alessandro's original results.

The number of test pairs differs between the two analyses for two reasons. I reserved 20% of the data for testing, compared with 30% in Alessandro's pipeline. For six of the eight pathways, this difference in proportion entirely explains the observed sample sizes. For the two largest excitatory pathways, however, the pipelines were not based on the same initial set of pairs, presumably because of additional subsampling or filtering, although the exact origin of this difference could not be to dd. This difference prevents a strictly pair-by-pair comparison, but does not call the main result into question: the curves obtained remain very close, as do the BCE and AUC values.

The most visible differences generally concern the least represented pathways, which is consistent with a more variable estimate of connection probability when fewer pairs are available.

2.6. Latent representations and connectome symmetries

With this first step complete, I moved on to studying a new research paper suggested by Professor Sanzeni. Adapting this method would form the basis of the remainder of this report and of my own contribution to the team's work. The more elementary model implemented previously nevertheless had to be presented, since it provides the foundations needed to understand what follows.

The paper is entitled *Graph Embeddings for Identifying Symmetries in Connectomes* and concerns the identification of symmetries and invariances in the fruit fly's internal compass. [13]

At first sight, the adjacency matrix of this circuit looks like an almost unreadable tangle. No obvious organisation emerges from it: the connections appear to be distributed almost at random.

However, by first grouping neurons according to type and then ordering them within each type according to their angular preference (this no longer refers to the orientation of a visual stimulus, but to the direction of the head for which their activity is highest), a previously invisible architecture appears.

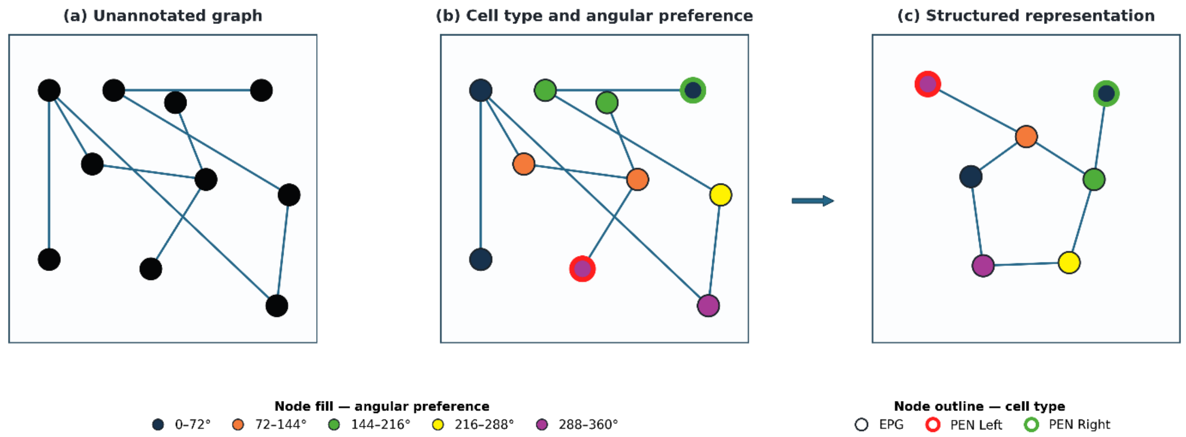


Figure 11 - Schematic identification of organisation within a neuronal graph. (a) Connectivity shown without annotations. (b) Addition of neurons' preferred heading, distributed between 0°and 360° and indicated by their colour, as well as their cell type: EPG neurons have no coloured outline, while PEN Left and PEN Right neurons are outlined in red and green respectively. (c) Reordering and grouping of neurons according to type and preferred heading, revealing a more structured pattern.

This organisation provides a better understanding of how the fruit fly's internal compass system operates.

2.6.1. The fruit fly's internal compass

A simple analogy is to compare this circuit to an incremental encoder linked to a circular counter. Imagine a disc divided into angular sectors covering the full 360°: the system retains an estimate of the direction of the head, then updates it with each rotation.

- **EPG neurons:** they act as a memory in which the direction of the head is represented on this disc. Each group of neurons corresponds to an angular sector and, at any given moment, only a few neighbouring neurons are highly active. This active region, known as the *activity bump*, indicates the direction currently held in memory. Thanks to recurrent connections between EPG neurons, this position can be maintained even in the temporary absence of a visual landmark.
- **Right and left PEN neurons:** they update this representation according to the direction in which the fly's head rotates. Their activation shifts the active region one way or the other around the disc, which amounts to incrementing or decrementing the remembered orientation.

The EPG population thus constitutes the circular register in which heading is represented, while the PEN populations update it on the basis of the fly's rotations.

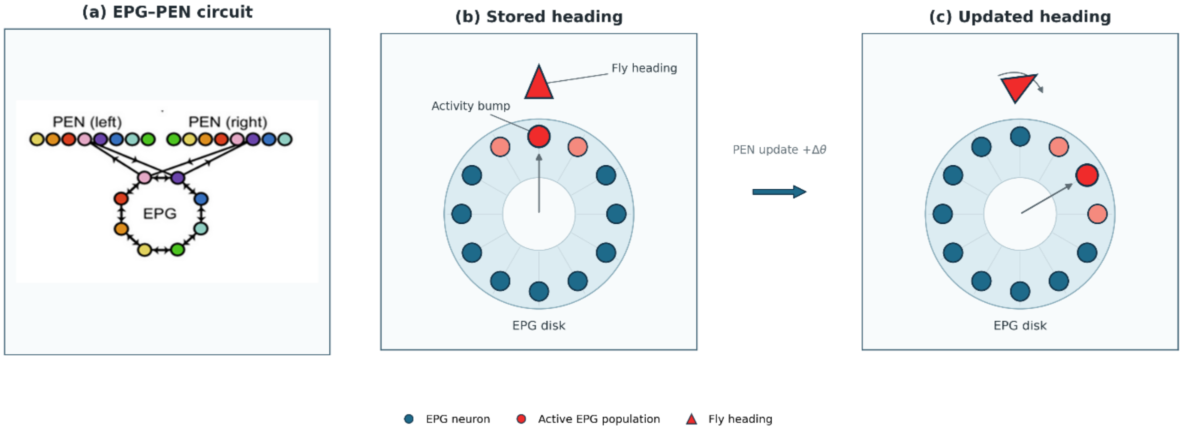


Figure 12 - EPG-PEN circuit and direction updating. (a) Idealised model of the EPG-PEN circuit, reproduced from Shan and Litwin-Kumar (2025). (b) The position of the active EPG population represents the remembered direction. (c) During a rotation, PEN neurons shift this activity in order to update the direction. [13]

This organisation exhibits rotational symmetry: if the angular preference of every neuron is shifted by the same amount, the neurons are permuted around the circle without changing the connectivity pattern. This is known as a circular graph.

This symmetry is reflected in the adjacency matrix. By first grouping neurons according to type, then ordering them within each type according to their angular preference, the matrix decomposes into circulant blocks: each row of a block is a cyclic shift of the preceding one.

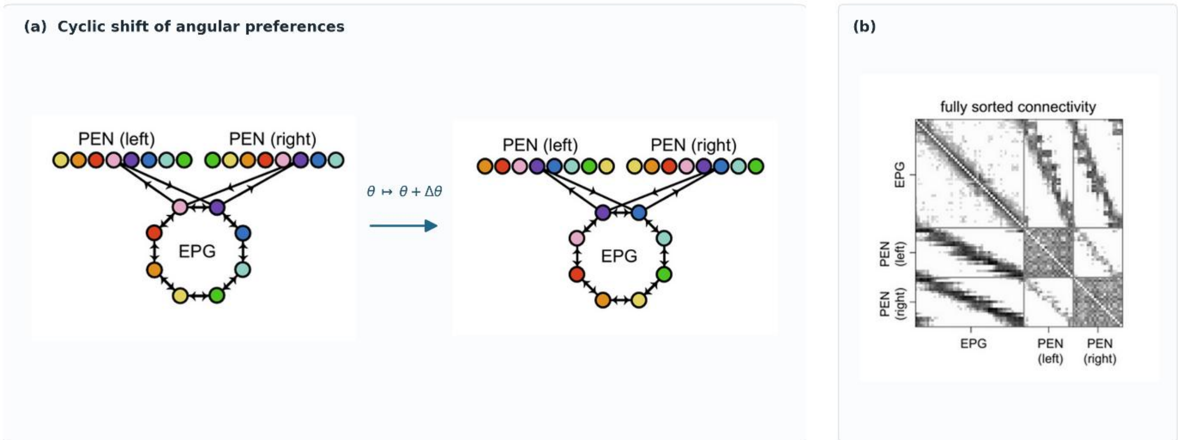


Figure 13 - Rotational symmetry of the EPG-PEN circuit. (a) A cyclic shift in angular preferences permutes the neurons without changing connectivity. (b) Sorting by neuronal type and then by angular preference reveals a block-circulant organisation in the adjacency matrix. Adapted from Shan and Litwin-Kumar (2025). [13]

In this example, the rotational symmetry reveals the existence of a continuous and periodic variable, whose meaning can then be identified from the biological data: the direction of the head.

However, this organisation became visible only after the neurons had been ordered using information known *a priori*, namely their type and angular preference. The challenge is therefore to recover these latent variables without having such annotations.

Can the encoded variable and its associated symmetries be made to emerge from the adjacency matrix alone?

2.6.2. Euclidean embedding model

This is where the paper proposes a surprising idea: assigning coordinates in a latent space of fixed dimensionality to each neuron. Initially random, these coordinates are progressively adjusted during training: the model thereby organises the neurons in this space so as to reproduce the observed adjacency matrix as closely as possible. If connectivity is structured by a hidden variable, that variable should then be reflected in the learned geometry.

The paper uses a decoding function based on the distance between latent representations to reconstruct synaptic weight:

$$\widehat{W}_{ij} = A_{c_i,c_j} \exp\left(-\frac{d(\boldsymbol{Z_i}, \boldsymbol{Z_j})^2}{\left(C_{c_i,c_j}\right)^2} \right) + B_{c_i,c_j}$$

Where $\widehat{W}_{ij}$ *is the predicted synaptic weight, that is, the number of synapses from neuron i to neuron j,* $\boldsymbol{Z_i}$ *and* $\boldsymbol{Z_j}$ *are their respective latent representations, and* A_{c_i,c_j}, B_{c_i,c_j} *and* C_{c_i,c_j} are learned parameters specific to each pair of neuronal types.

A *softplus* is added to the decoder output in order to define a positive parameter:

$$\widetilde{\mu_{ij}} = \mathrm{softplus}(\widehat{W}_{ij}) = \log(1 + \exp(\widehat{W}_{ij}))$$

We then assume that the observed number of synapses follows a Poisson distribution with parameter μ_{ij}.

$$W_{ij} \sim \mathrm{Poisson}(\mu_{ij}).$$

The loss function then quite naturally becomes the negative Poisson log-likelihood, supplemented by a regularisation term:

$$\mathcal{L}(\boldsymbol{Z}, \boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}) = -\sum_{i \neq j} \log P_{\mathrm{Poisson}}\left(W_{ij} \mid \mu_{ij}\right) + \frac{\lambda}{ND} \|\boldsymbol{Z}\|_F^2$$

Adding the types explicitly through parameters A, B and C prevents the model from encoding them in latent space and allows that space to reveal finer structures, such as the symmetries being sought.

From these expressions, it is clear that the model adjusts the latent representations so as to bring strongly connected neurons closer together and move weakly connected or unconnected neurons further apart.

Finally, to determine whether the latent representations form a circular structure, the authors seek to match them to the points of a reference circle:

$$\mathcal{L}(T, Z) = \min_{\Pi, P} \|T - \Pi Z P\|^2$$

The matrix $Z \in \mathbb{R}^{N \times D}$ contains the latent representations of the N neurons

The matrix $T \in \mathbb{R}^{N \times 2}$ represents a reference circle composed of N evenly spaced points:

$$T_i = (\cos\theta_i, \sin\theta_i), \qquad \theta_i = \frac{2\pi i}{N}.$$

Since the representations Z_i have D dimensions, the matrix $P \in \mathbb{R}^{D\times 2}$ makes it possible to project them onto a plane. The product ZP therefore provides a two-dimensional representation of each neuron.

However, no angular position is initially assigned to the neurons. The permutation matrix $\Pi \in \mathbb{R}^{N\times N}$ therefore searches for the neuron ordering that best matches the order of the points on the circle.

The minimisation thus simultaneously searches for the projection P and ordering Π that provide the best match to a circle. A low value of $\mathcal{L}(T, Z)$ therefore reveals the presence of a circular organisation in latent space.

Unlike the adjacency matrix in the paper, which is weighted by the number of synapses, ours indicates only the presence or absence of a connection. The model must therefore be adapted. We retain the proposed reconstruction function, but transform its output into a probability using a sigmoid. The BCE then compares these probabilities with the observed binary values and thus forms our loss function.

$$\tilde{p}_{ij} = \mathrm{sigmoid}\left(A_{c_i,c_j} \exp\left(-\frac{d(Z_i, Z_j)^2}{C^2_{c_i,c_j}}\right) + B_{c_i,c_j}\right)$$

And

$$\mathcal{L}_{ij} = \mathrm{BCE}(Y_{ij}, \tilde{p}_{ij})$$

In the paper, this model works particularly well on the fruit fly's internal compass: it recovers the symmetry described above and, consequently, the variable encoded by this circuit.

In contrast, applying it to the mouse visual connectome produces less conclusive results. This outcome was expected: unlike the fruit fly's internal compass, the subregion studied within primary visual area V1 does not serve to encode one particular variable - or at least not at this scale.

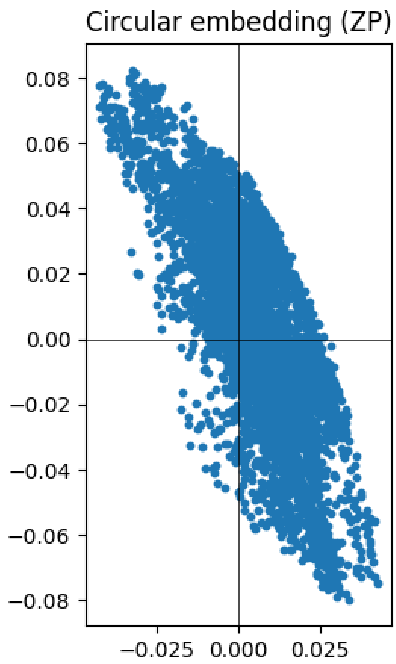


Figure 14 - Search for a circular organisation in the latent space of the $E_{L2/3} \rightarrow E_{L2/3}$ pathway. Each point corresponds to the two-dimensional projection optimised to obtain a circle ($Z_i P$). Despite this optimisation, the representations form an elongated cloud rather than a ring and therefore reveal no clear circular organisation.

3. Adapting the method to the MICrONS connectome

These results led us to change perspective. The objective was no longer to recover a known symmetry, but to use latent space as a tool for exploring connectivity. Two questions then guided the rest of my work: how many latent dimensions are needed to account for our adjacency matrix as accurately as possible? And once this number has been determined, what do the learned latent variables represent: do they correspond to biological properties that have already been measured, or do they reveal a previously unknown organisation?

3.1. Adapting the decoder to binary connections

Since our aim is no longer to order neurons to reveal a particular symmetry, the geometry of latent space is no longer as important. Alongside the model proposed in the paper, which we will describe as "Euclidean" because it relies explicitly on the distance between Z_i and Z_j, we therefore introduce a more flexible model:

$$\tilde{p}_{ij} = \mathrm{MLP}_\theta\left(Z_i, Z_j\right)$$

The MLP estimates the connection probability between two neurons directly from their latent representations. The vectors Z_i therefore play a dual role: they are both parameters learned by the model and the inputs to the MLP. During training, the latent representations and the network parameters θ are jointly optimised to explain the adjacency matrix as accurately as possible.

We therefore replace not the loss function, which remains the BCE, but the Euclidean decoder with an MLP. This modification makes the model more flexible, at the cost of a latent space that is less directly interpretable and consequently closer to a genuine black box.

3.2. Training on the verified ("proofread") subset

To limit computational cost, we focus our analysis on the $\left(E_{L2/3} \rightarrow E_{L2/3}\right)$ connections, the most highly represented in the connectome. We retain only *proofread* neurons, that is, neurons whose reconstruction has been checked and, where necessary, manually corrected in order to limit segmentation errors.

The first trials are quite disconcerting:

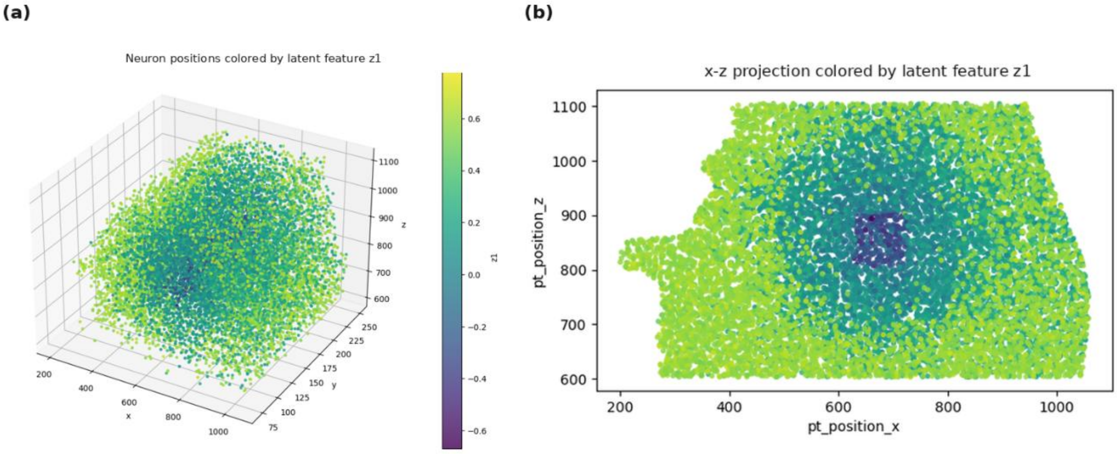


Figure 15 - Spatial organisation of the latent representations learned by the MLP model. (a) Three-dimensional view of neuronal positions; (b) projection onto the $(x; z)$ plane. Colour represents Z_1 for the model with three latent dimensions. Lower-dimensional models exhibit qualitatively similar spatial organisations.

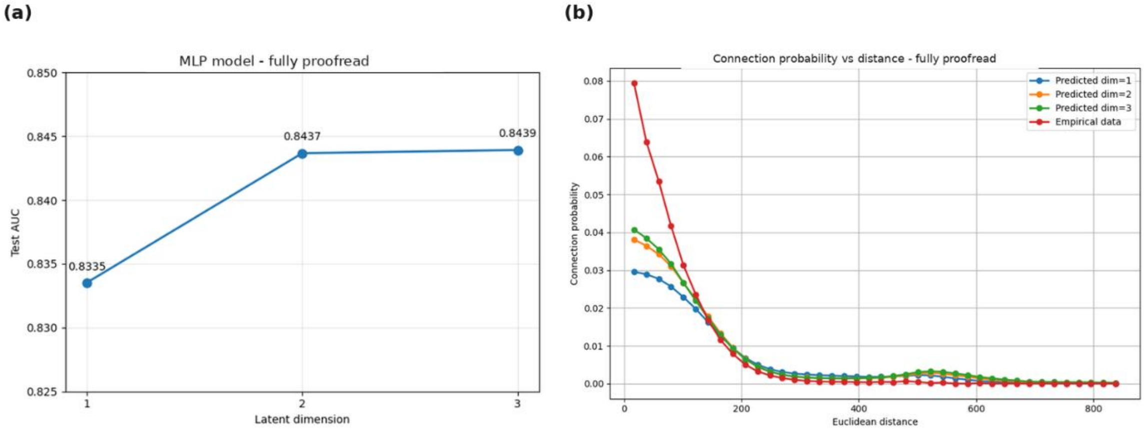


Figure 16 - Influence of latent dimensionality on MLP performance. (a) AUC on the test set as a function of the number of latent dimensions; (b) predicted connection probabilities as a function of Euclidean distance, compared with the empirical data

3.3. Identifying and controlling for a reconstruction bias

Alessandro's geometric model, which estimated connection probability from intersomatic distance, achieved an AUC of $0{,}82$ for $E_{L2/3} \rightarrow E_{L2/3}$ connections. Yet with a single latent coordinate per neuron, compared with three spatial coordinates for the geometric model, our MLP achieved the same level of performance, with an AUC of 0,83. Mapping each neuron's latent value to its anatomical position revealed a particular organisation: this value appeared to depend on its distance from the centre of the connectome.

My first hypothesis was a boundary effect. Peripheral neurons have fewer neighbours and therefore fewer opportunities to form connections than central neurons, which mechanically reduces their number of observable connections. This interpretation was consistent with in-degree and out-degree, which were higher at the centre of the volume and almost linearly correlated with the latent variable.

Professor Sanzeni nevertheless proposed a second explanation, related to the *proofreading* process. The neurons were probably not all reconstructed with the same level of accuracy. Since tracing axons and dendrites over long distances is particularly difficult, central neurons may have been reconstructed more completely than peripheral neurons. The larger number of visible connections at the centre could therefore arise from a reconstruction bias rather than a biological phenomenon.

To distinguish between these two hypotheses, we spatially divided the connectome and then retrained the model on the resulting subvolumes. If the latent representation arose solely from a boundary effect, neurons that were initially central but became peripheral after the split should have changed value. Yet their representations remained approximately unchanged: the new plots essentially corresponded to half of the initial plot. This result therefore rules out the hypothesis of a simple boundary effect and supports that of a spatial bias related to *proofreading*.

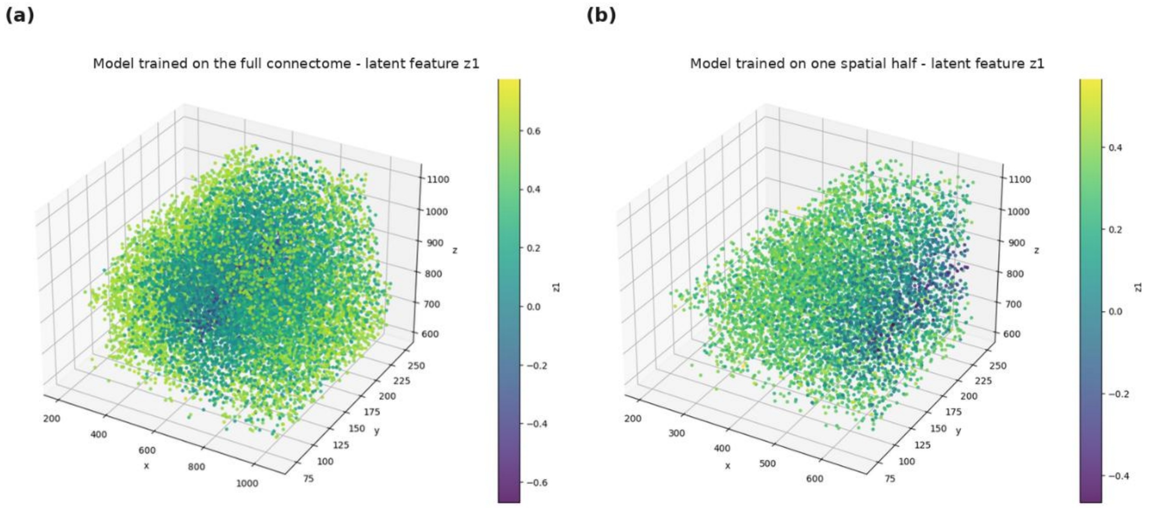


Figure 17 - Preservation of the spatial organisation of the latent representation after splitting the connectome. Colours represent the first latent coordinate Z_1 of the three-dimensional MLP model: (a) model trained on the entire connectome; (b) model retrained independently on one spatial half of it. The profile observed in the subvolume remains qualitatively similar to the corresponding portion of the initial profile. Since the colour scales are specific to each training run, the comparison concerns the spatial organisation of the values rather than their absolute magnitude.

3.4. Extension to an unverified ("non-proofread") region

But how could we get rid of this bias? One natural idea is to perform the calculations on the entire connectome rather than on the *proofread* subset alone.

3.4.1. Computational constraints and strategies considered

Extending this analysis to the entire connectome nevertheless creates an immediate computational difficulty. The complete connectome contains approximately $13{,}157$ excitatory $E_{L2/3}$ neurons. Considering every possible ordered pair gives:

$$13{,}157 \times 13{,}157 \approx 173 \text{ million potential connections}$$

Materialising all these pairs requires storing, for each of them, the identifiers of the two neurons, its label and various related pieces of information. This represents several gigabytes with compact tensors, and more than 15 GB when the data are handled as Python objects. Such a volume was becoming difficult to process on my computer, notably because of its VRAM.

At this stage, three solutions seemed possible to me:

- **Subsample the negative examples:** the drawback is that we partly lose the true biological proportion between connected and unconnected pairs. This therefore no longer allows the model to be evaluated by comparing predicted connection probability with empirical probability as a function of Euclidean distance, as we did previously.
- **Change the observation window:** we analyse another part of the connectome whose shape is similar to that of the *proofread* volume. This approach nevertheless raises two main difficulties: ensuring that the regions compared have a homogeneous composition in terms of reconstruction levels, and having a sufficient number of neurons.
- **Use CINECA's computing resources:** a supercomputer accessible through the Politecnico di Milano cluster.

Given the time available and the cumbersome procedure for gaining access to CINECA, we initially focused on the second option. If it proved conclusive, the analysis could then be extended to the entire connectome using CINECA's resources.

3.4.2. Selecting a new analysis window

The *proofread* portion of the connectome lies at the centre of the volume. Since the observed bias might stem from the inhomogeneity of this manual reconstruction, it seemed worthwhile to move the analysis window to a region without *proofread* neurons. In this area, the reconstruction produced by automatic segmentation probably remains imperfect over long distances, but its error level should be more homogeneous. We therefore selected the lower-left quarter of the connectome.

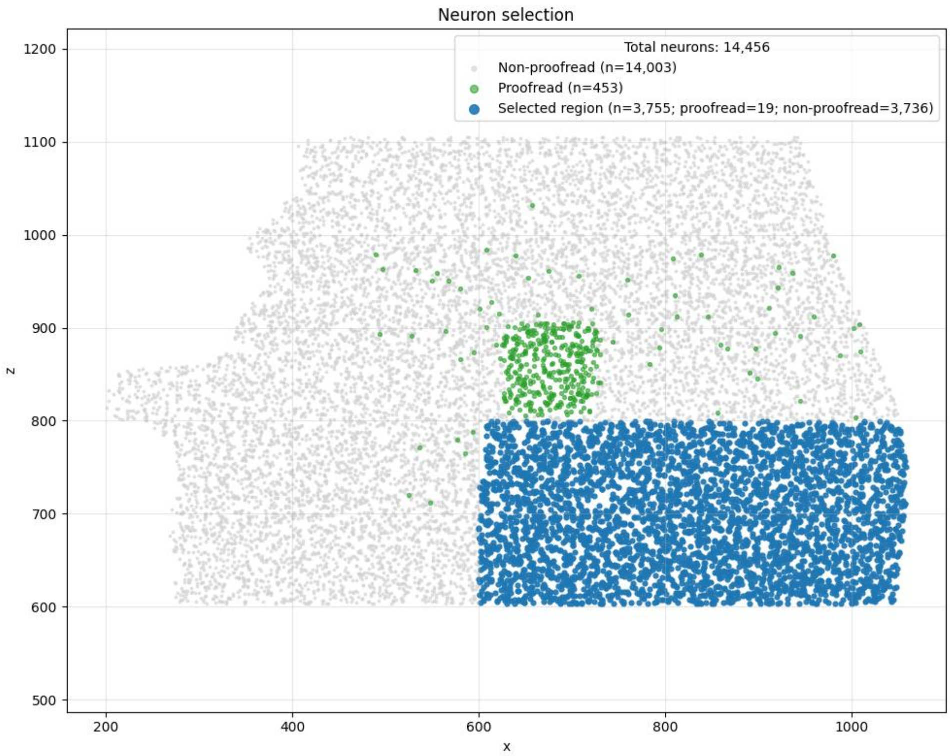


Figure 18 - Selection of the new analysis window. Projection of the 14,456 neurons onto the (x; z) plane. In the selected region, 99.5% of the neurons come from automatic segmentation.

The results are then far more consistent: connection probability as a function of intersomatic distance becomes much closer to the empirical data. And we managed to get rid of this centre-dependent structure.

4. Results and interpretation of latent representations

4.1. Performance and latent-space dimensionality

Before analysing the results, let us clarify what we expect to observe. Our hypothesis is that connectivity depends strongly on anatomical distance: two nearby neurons are more likely to be connected. If this hypothesis is correct, moving from one to two or three latent dimensions should substantially improve performance, since these additional dimensions could represent the spatial coordinates that are absent at lower dimensionality.

This is precisely what we observe with the MLP, whose performance quickly reaches a plateau at around two or three dimensions. In contrast, the model from the paper begins improving later and continues to improve up to approximately eight dimensions. Below three dimensions, its performance is close to random ranking, whereas the MLP already achieves an AUC of $0{,}77$ with a single dimension.

The most interesting result nevertheless appears when we compare the latent MLP with a simple geometric model. We trained an MLP using only the anatomical coordinates of the neurons in the portion of the connectome studied: $(x_i, y_i, z_i, x_j, y_j, z_j)$.

This model achieves an AUC of approximately $0{,}86$, compared with $0{,}91$ for the latent MLP. A gain of $0{,}05$ AUC is substantial at this level of performance. The latent representations therefore capture information that is not directly contained in the anatomical coordinates alone.

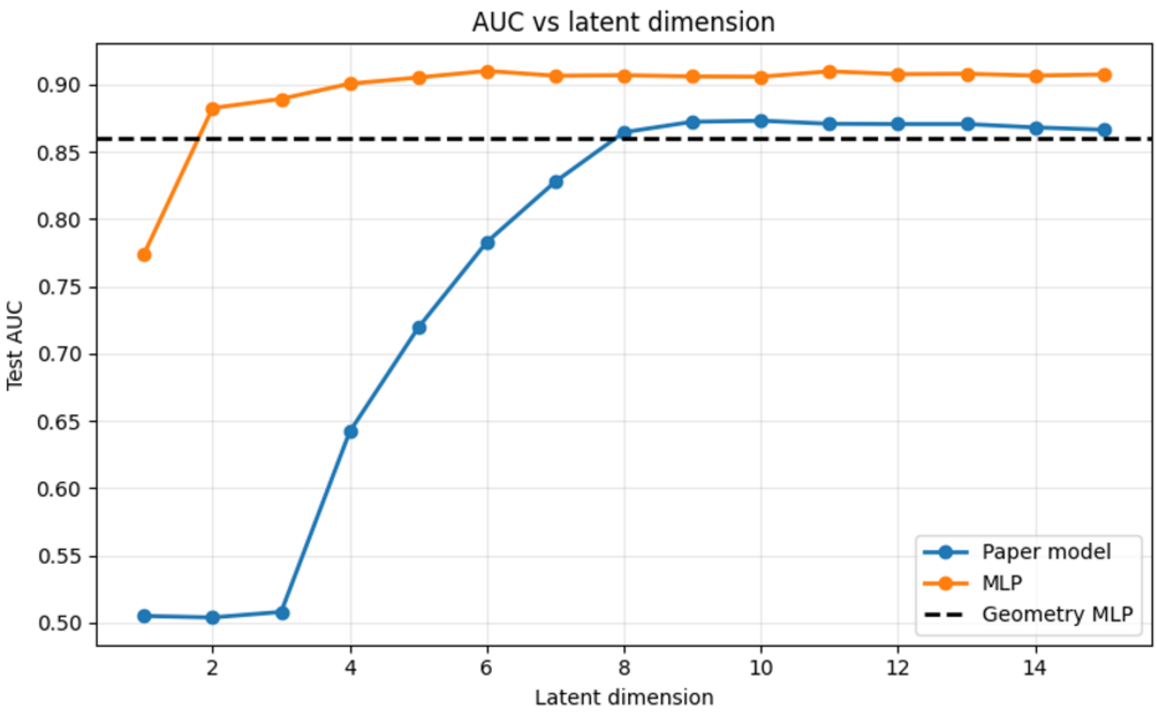


Figure 19 - Influence of latent dimensionality on model performance. AUC on the test set as a function of the number of latent dimensions, within the selected region of the connectome.

What, then, do these additional dimensions represent, and why do they improve predictions so substantially?

Let us take a closer look at this latent MLP model:

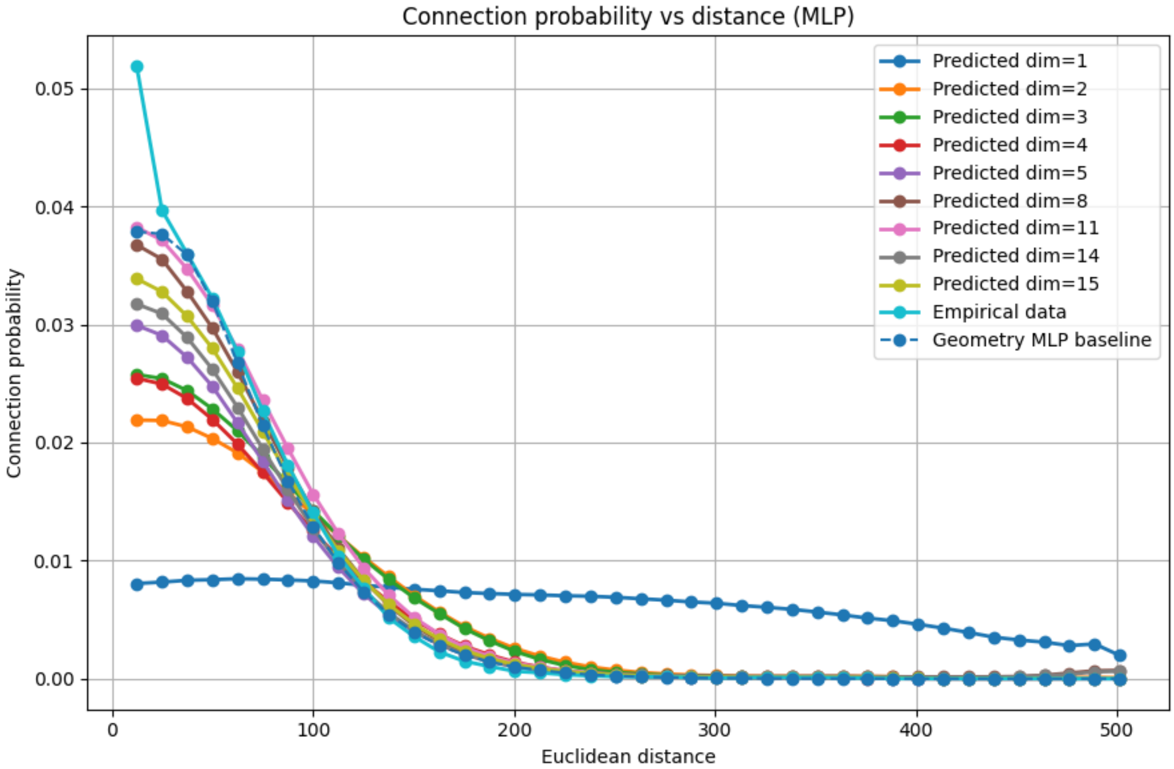


Figure 20 - Connection probability as a function of anatomical distance. Comparison of the empirical and predicted probabilities from the different models.

As the number of dimensions increases, the predicted curve progressively approaches the empirical curve, up to approximately eleven dimensions. Beyond that point, improvements become minimal and the variations observed appear more likely to reflect fluctuations than a systematic gain.

One observation nevertheless remains intriguing: with a single latent dimension, the mean predicted probability remains almost constant regardless of distance, even though the model achieves an AUC close to 0,77. These two results are not necessarily contradictory: the curve represents a mean for each distance interval, whereas the AUC evaluates the ranking of individual pairs. The model may therefore distinguish certain pairs within the same interval without reproducing the average dependence on distance. It nevertheless remains to be determined precisely what information it uses to do so.

4.2. Geometry recovered in latent space

The relationship between latent and anatomical distance is particularly clear. Yet we never asked the model to calculate Euclidean distances or preserve the geometry of the connectome. It therefore discovers spontaneously that anatomical distance is one of the main determinants of neuronal connectivity.

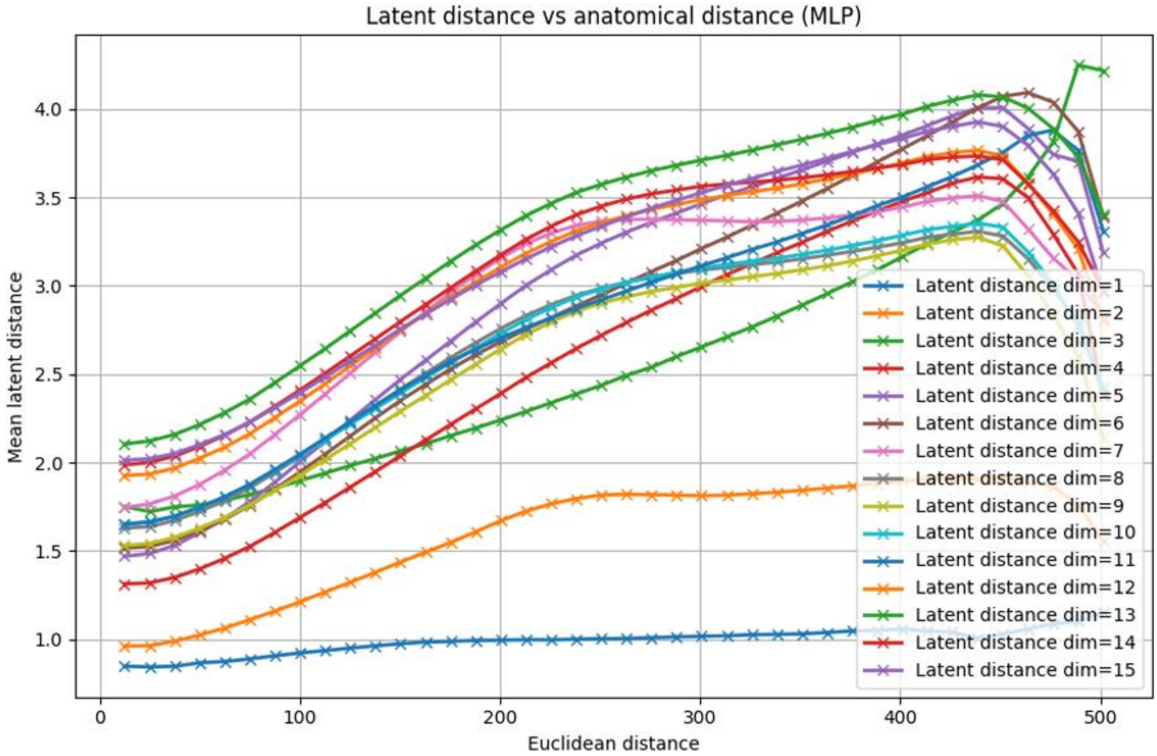


Figure 21 - Relationship between latent and anatomical distances. Mean latent distance between pairs of neurons as a function of their anatomical distance, for model latent spaces with dimensionalities from 1 to 15.

Qualitatively, the model does appear to encode the spatial variables. This is demonstrated by the colour gradients that appear when we plot the positions of the neurons and colour them according to the value of a latent variable.

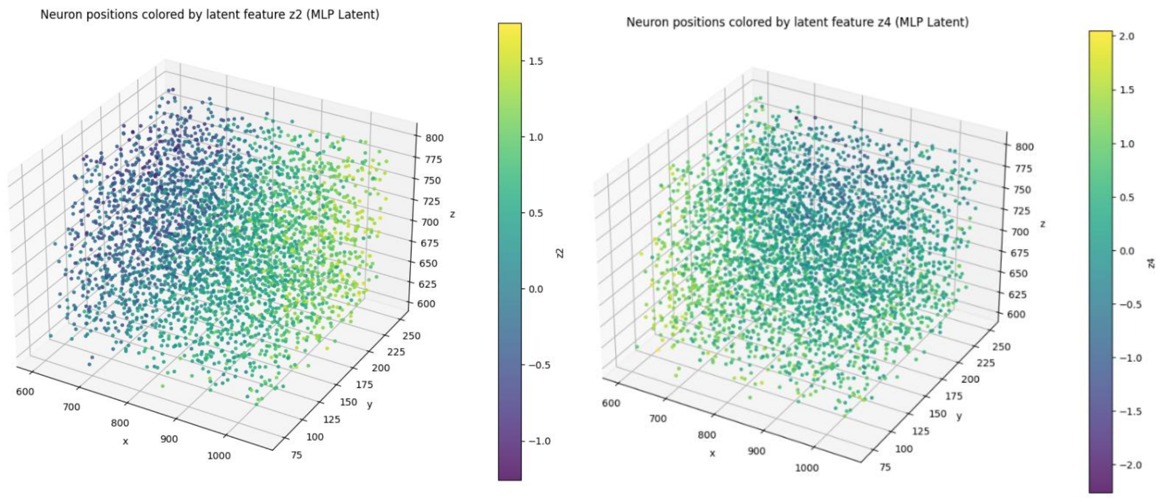


Figure 22 - Spatial organisation of the latent coordinates. Anatomical positions of the neurons coloured according to (Z_2) (a) and Z_4 (b), learned by the 11-dimensional MLP model.

This observation is confirmed when we examine the correlations. I calculated Spearman correlations between the latent and biological variables for the eleven-dimensional model, which achieves a high AUC and most faithfully reproduces the empirical relationship between distance and connection probability.

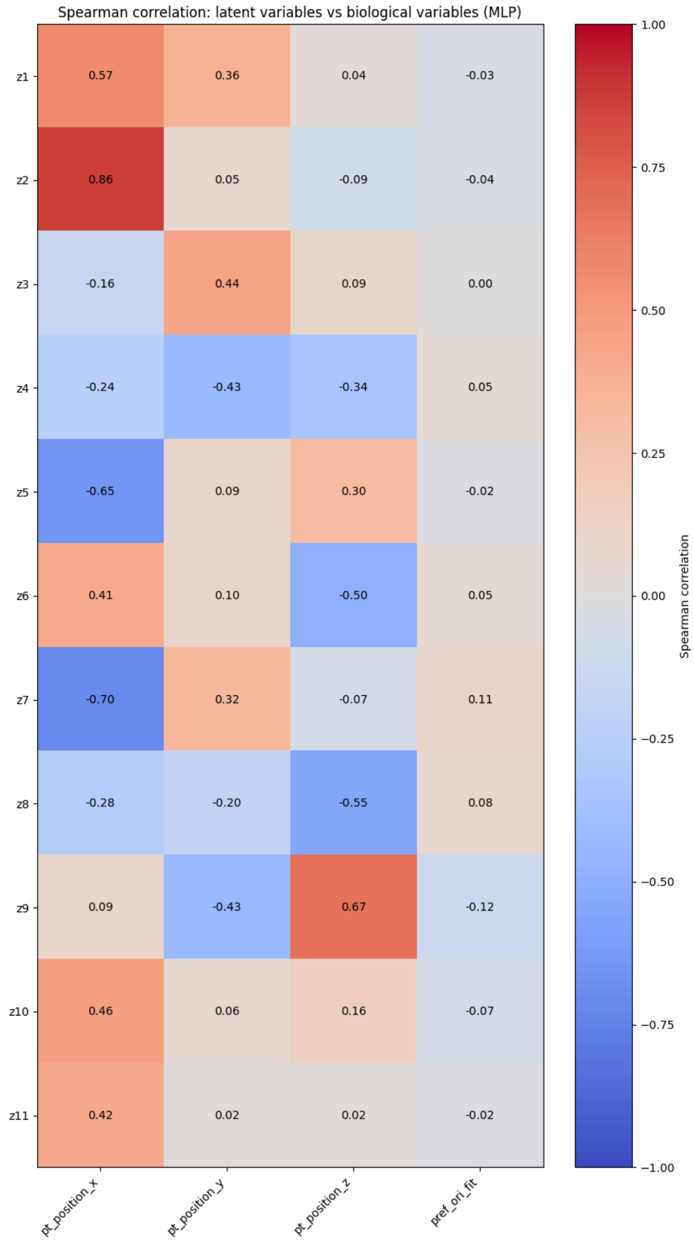


Figure 23 - Correlations between latent variables and neuronal properties. Spearman correlation coefficients between the 11 latent dimensions of the MLP, the anatomical coordinates of the neurons and their angular preference.

For almost every latent variable, we observe a non-negligible correlation with at least one spatial coordinate, sometimes x, sometimes y or z. In contrast, the latent variables do not appear to be related to the neurons' orientation preference.

This observation is also visible when we project latent space onto its first two principal components using PCA, then colour the neurons according to the different biological variables studied.

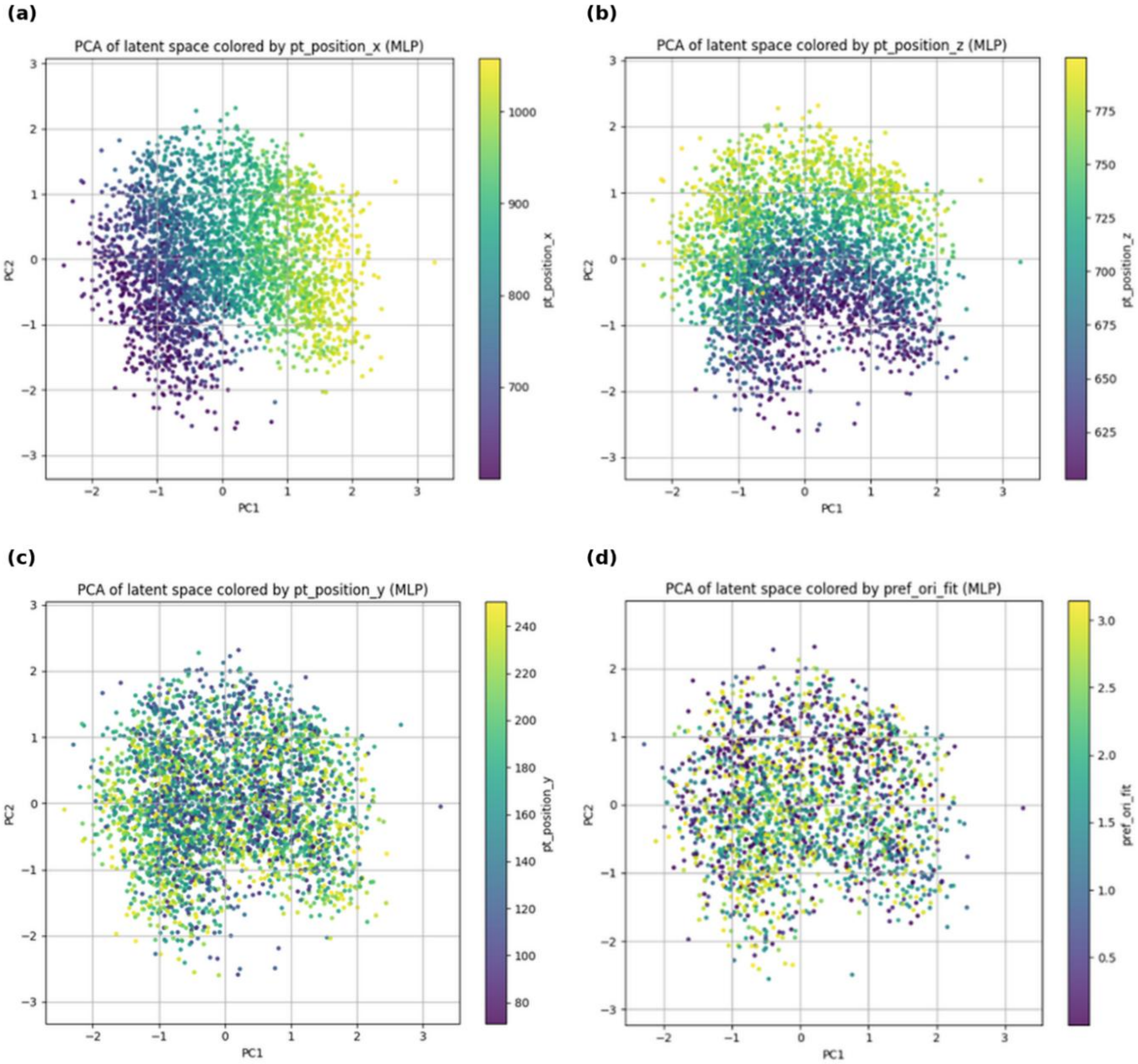


Figure 24 - PCA projection of latent space. Projection of the 11-dimensional MLP representations onto their first two principal components, coloured according to anatomical positions x (a), z (b) and y (c), then according to angular preference (d).

The colour gradients observed once again confirm the close relationship between the spatial coordinates x and z and the latent variables. Conversely, the scattered distribution of the colours associated with orientation preference indicates that even if it is encoded to a small extent, it is clearly not predominant in latent space.

The conclusion of this section is particularly strong. From the adjacency matrix alone, without providing the MLP with any anatomical or functional information, the latent representations spontaneously recover the main result of Alessandro's thesis: spatial proximity is the principal determinant of neuronal connectivity, while orientation preference, at the heart of the *like-to-like* hypothesis, appears to play a more secondary role.

The Euclidean model from the paper, when applied to our connectome, leads to observations very similar to those of the latent MLP. This convergence is consistent, since the MLP also learns to represent distance even though distance is never explicitly provided to it. The results of the Euclidean model nevertheless remain less clear-cut, suggesting that the MLP's greater flexibility allows it to capture additional information beyond spatial proximity alone. Which remains for us to determine!

4.3. Contribution of functional variables

Given the consistency of these observations, identifying the source of the latent MLP's AUC gain over the geometric model was a logical and particularly promising next step in our analysis.

But how can we identify the nature of this additional information? The latent MLP is a genuine black box: it assigns each neuron a representation whose meaning we do not know, then itself learns how to combine two of these representations to estimate their connection probability. We therefore know neither what latent space encodes nor the rule by which the model interprets it: two unknowns.

To address this problem, we chose to fix the inputs while retaining the MLP decoder. The learned latent representations are replaced by known biological variables, introduced individually and then in combination, and each model is trained to reconstruct the adjacency matrix. The change in AUC then makes it possible to evaluate the predictive power contributed by each variable. If a combination recovers the performance of the latent MLP, this suggests that it contains a substantial proportion of the information encoded by the representations.

But which other biological variables should we test? Anatomical position is already taken into account by the geometric model, while Alessandro's thesis and our previous analyses show that orientation preference has limited discriminative power. We must therefore look for other properties that might explain differences in connectivity between neurons.

4.3.1. Functional data and matching

As we saw in the introduction, MICrONS provides not only an anatomical reconstruction of the connectome, but also functional recordings of neuronal activity. Until now, our models have directly exploited only the anatomical component of the data. The functional component has been used only occasionally, through angular preference, without ever being provided as an input to the model.

The protocol for acquiring the functional data was based on presenting the mouse with three broad categories of videos:

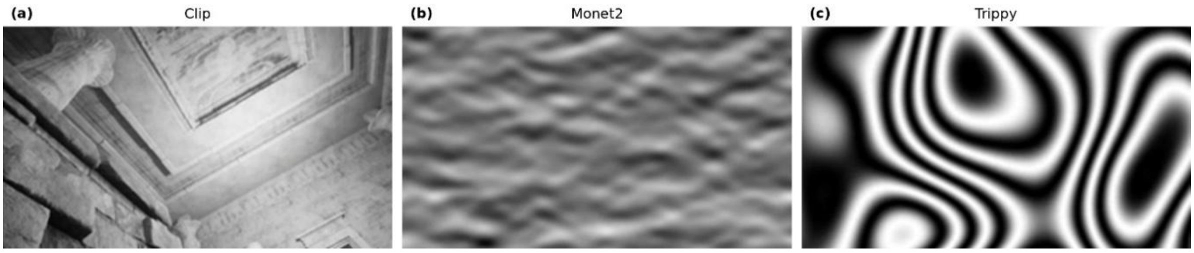


Figure 25 - Visual stimuli used for functional data acquisition. The *Clip* stimulus represents a camera moving through a visual environment (a), *Monet2* combines Gaussian noise smoothed according to an overall orientation and direction (b), while *Trippy* varies these properties locally through irregular dynamic bands (c)

Each frame of the video is divided into patches. At each time point (t), each patch is described by a vector of 512 features (shapes, orientations, spatial frequencies...)

$\mathbf{h}_{i,j}^{t} \in \mathbb{R}^{512}$ known as a *feature vector*

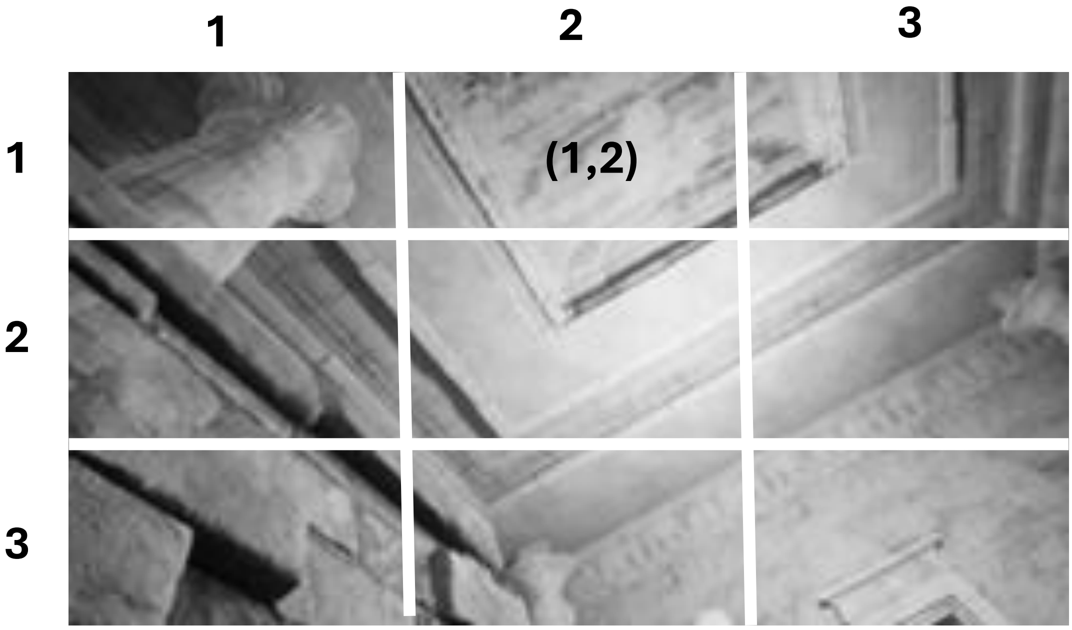


Figure 26 - Spatial division of the images and feature extraction.

A model is trained on all $\mathbf{h}_t(i,j)$ to determine which patch, defined by its *readout position* (μ_x, μ_y), and which of the 512 variables best predict neuronal activity.

$$(\hat{\mu}_{x,n}, \hat{\mu}_{y,n}, \hat{\boldsymbol{w}}_n) = \underset{\mu_x, \mu_y, \boldsymbol{w}}{arg\,min} \sum_t \left[\boldsymbol{w}^\top \boldsymbol{h}_t(\mu_x, \mu_y) - r_n(t)\right]^2$$

The measured activity of neuron n at time t is represented by $r_n(t)$. The readout feature-weights vector $\boldsymbol{w}_n \in \mathbb{R}^{512}$ indicates the importance, or weight, of each of the 512 visual features in predicting its activity. Finally, the pair $(\mu_x, \mu_y)_n$ defines the readout position corresponding to the patch of the video whose features best predict its activity. This position is known as the receptive field centre (RF centre).

Without dwelling on this technical aspect, this information is distributed across three tables that form a matching chain. The first contains the feature weights $\boldsymbol{w}_n$ and the *readout position* $(\mu_x, \mu_y)_n$ of each recorded unit, but no neuronal identifier. The second lists these units in the same order for each session and scan, which makes it possible to assign them a unit_id. The triplet (session, scan, unit_id) then uniquely identifies a functional unit. A third table finally establishes the link with anatomy by associating this triplet with the pt_root_id used for the anatomical component in the adjacency matrix.

Since this final registration is automatic, each match is evaluated using two criteria. The residual error measures the distance between the registered functional position and the selected anatomical soma, while the separation score compares this distance with that of the nearest competing soma.

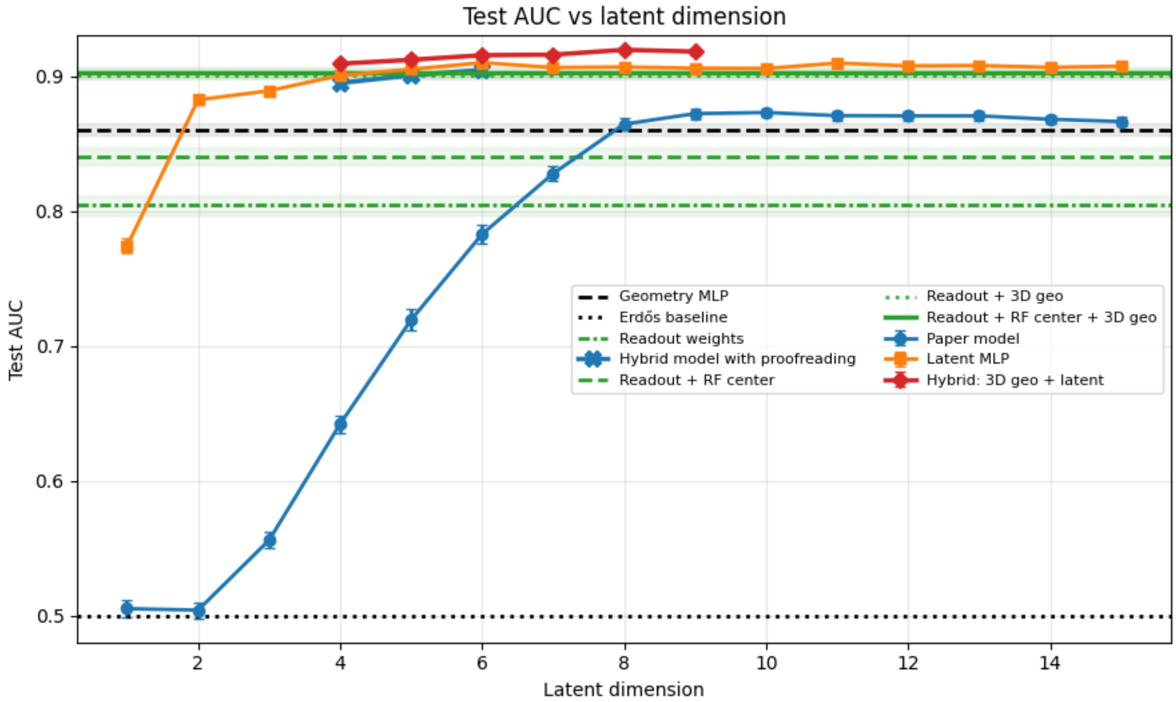


Figure 27 - Comparison of anatomical, functional and latent model performance. AUC on the test set is shown as a function of the dimensionality of the latent models. The curves correspond to the model from the paper (blue), the latent MLP (orange) and the hybrid model combining anatomical coordinates and latent variables (red). The horizontal lines indicate the performance of models based on geometry alone (black) or on different combinations of functional variables (green).

This graph is undoubtedly the main culmination of the first three months of this internship. Behind this figure lie several hundred hours of computation, but above all the entire line of reasoning developed thus far. The geometric model had shown that anatomical position explained a large part of connectivity, without nevertheless matching the performance of the latent MLP. Yet, when we add the *feature weight* and the receptive-field centre (*RF centre*) to the three-dimensional coordinates, the AUC almost exactly matches that of the latent model.

For the first time, this result suggests a possible biological interpretation of the model's predictive gain. The measured anatomical and functional variables are sufficient to reproduce almost all the performance of the representations learned from the adjacency matrix alone. In other words, the black box appears to have recovered by itself not only the position of the neurons, but also some of their functional properties. The gap between the geometric model and the latent MLP would therefore not arise from an inexplicable structure: it could correspond to the functional information encoded by the feature weights $\boldsymbol{w}_n$ and the *readout position* $\left(\mu_x, \mu_y\right)_n$

To test this hypothesis, we must now extend our various tools for analysing latent space to the functional data.

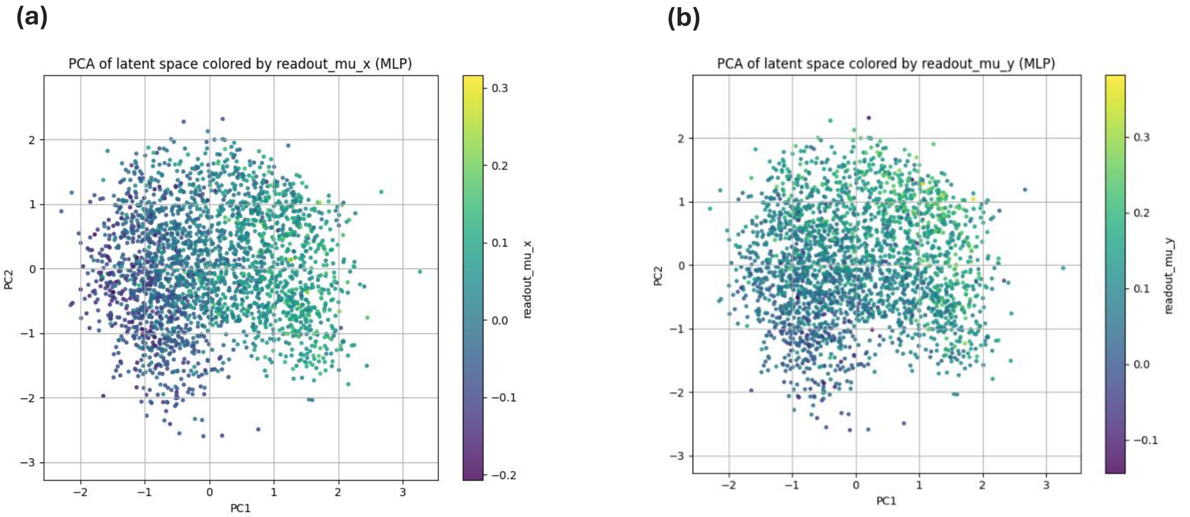


Figure 28 - PCA projection of latent space as a function of readout position. Projection of the 11-dimensional MLP representations onto their first two principal components, coloured according to μ_x (a) and μ_y (b)

PCA once again reveals colour gradients: mainly along the first component for μ_x, and less markedly for μ_y. Their organisation strongly recalls that observed previously with the anatomical coordinates x and z. This suggests a close relationship between the position of neurons in the cortex and that of their receptive field in visual space, although a quantitative analysis is still needed to measure these correlations precisely.

We thereby recover a fundamental property of visual area V1: its retinotopic organisation. Neighbouring regions of the visual field are mainly processed by populations of anatomically neighbouring neurons. This organisation therefore appears to be reflected in the latent space learned from the connectivity of $E_{L2/3}$

This redundancy nevertheless suggests that the additional information responsible for the AUC gain lies not primarily in the receptive-field centre, but rather in the *readout feature weights*.

4.3.2. Decoding latent representations

The *readout feature weights* do not, however, form a scalar variable, but rather a vector with 512 components. The correlations and visualisations used thus far, which examine each variable separately, are therefore not well suited. The functional information sought may be distributed across several components and become visible only through a linear, or even non-linear, combination, without any one of them individually exhibiting a marked correlation with latent space. A multivariate analysis method therefore becomes necessary.

To determine whether the functional data can recover latent space, we associate with each neuron (i) its feature weights $\boldsymbol{w}_i$, which we will denote $\mathbf{R}_i \in \mathbb{R}^{512}$ to avoid any confusion with the model's other parameters, and its latent representation $\mathbf{Z}_i \in \mathbb{R}^{11}$. Two decoding functions are compared:

$$\hat{\mathbf{Z}}_i^{\text{linear}} = \mathbf{W}\mathbf{R}_i + \mathbf{b}, \qquad \hat{\mathbf{Z}}_i^{\text{MLP}} = \text{MLP}_\theta(\mathbf{R}_i)\,.$$

Their parameters are learned on the training neurons by minimising the regularised squared error:

$$\mathcal{L}_{\text{dec}}(\theta) = \frac{1}{N_{\text{train}}} \sum_{i \in \mathcal{D}_{\text{train}}} \left|\mathbf{Z}_i - \hat{\mathbf{Z}}_i\right|_2^2 + \lambda |\theta|_2^2$$

Predictive power is then measured by the R^2 coefficient on neurons held out from training. It is compared with a control in which the latent representations are randomly permuted, $\mathbf{R}_i \mapsto \mathbf{Z}_{\pi(i)}$.

Decoding latent space from the readout feature weights

(a) Overall performance on the test set

Model	Test R²	Explained variance	Mean Euclidean distance	Median Euclidean distance	Mean cosine similarity	Median cosine similarity
Linear	0.1479	0.1502	2.9484	2.8872	0.3636	0.4369
Linear — permuted control	-0.1076	-0.1047	3.3680	3.3371	-0.0370	-0.0396
MLP	-0.0352	-0.0307	3.2643	3.2235	0.2699	0.3269
MLP — permuted control	-0.2759	-0.2721	3.6216	3.5832	0.0119	-0.0043

(b) R² coefficient on the test set for each latent dimension

Latent dimension	Linear	Linear — permuted control	MLP	MLP — permuted control
z_1	0.1636	-0.1088	-0.0799	-0.2102
z_2	0.3617	-0.1540	0.2489	-0.2774
z_3	-0.0179	-0.1535	-0.2267	-0.2370
z_4	0.0742	-0.0588	-0.1312	-0.2626
z_5	0.2264	-0.1111	-0.0987	-0.3399
z_6	0.0755	-0.0695	-0.1556	-0.2936
z_7	0.3000	-0.1285	0.1875	-0.3207
z_8	0.1045	-0.0500	-0.0728	-0.1975
z_9	0.1443	-0.0927	0.0456	-0.2645
z_{10}	0.1248	-0.1118	-0.0581	-0.3402
z_{11}	0.0694	-0.1452	-0.0458	-0.2910

Green cells: R² > 0 • Red cells: R² < 0

Figure 29 - Decoding latent space from the *readout feature weights*. (a) Overall performance of the linear and MLP decoders on the test set; (b) R^2 coefficient obtained separately for each of the eleven latent dimensions.

After retaining only neurons whose functional activity is predicted sufficiently well ($\text{cc}_{\text{abs}} > 0{,}2$) and excluding uncertain matches between the functional and anatomical data, performance remains modest.

Linear regression explains 15% of the variance in latent space. In contrast, the MLP obtains a negative R^2 on neurons held out from training: it is therefore unable to predict their latent representation more accurately than a simple prediction based on the mean.

4.4. Hybrid model and residual information

The preceding analyses show that geometry constitutes the dominant information in the representations learned by the MLP. This predominance could obscure the other connectivity factors we are seeking to identify. By providing the neuronal positions directly to the model, we therefore hope to free latent space from the need to encode this information and to reveal more clearly the properties that have so far been drowned out by the geometric signal.

To do so, we introduce a new model, which we call the **hybrid model**. It retains the architecture of the latent MLP but also receives as input the three-dimensional coordinates of the presynaptic and postsynaptic neurons. Since geometry is explicitly available to the decoder, the latent variables can focus on the residual information that escapes the purely geometric model.

As soon as a single latent dimension is added, the hybrid model achieves performance comparable to that of the complete latent MLP **(Figure 27).** Providing the coordinates does not, however, guarantee that the model will cease encoding them redundantly in latent space. We therefore check that positional information is indeed absent from it:

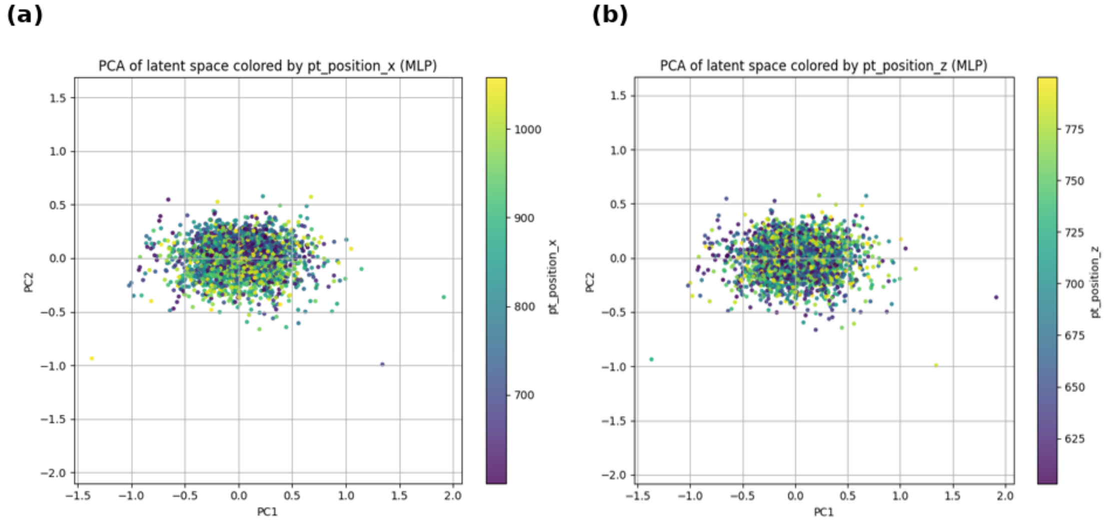


Figure 30 - Search for residual positional information in the latent space of the hybrid model. Projection of the latent representations onto the first two principal components, coloured according to anatomical coordinate x (a) and anatomical coordinate z (b).

Since adding the single latent variable Z_1, which we have verified no longer contains positional information, to the three-dimensional coordinates is sufficient to recover almost all the performance of the latent MLP, we conclude that Z_1 contains most of the information responsible for the AUC gain over the purely geometric model. Identifying the nature of this information therefore becomes the core of the analysis.

It appears that the latent representation still does not encode the functional data

Decoding the residual latent variable from readout feature weights

Test-set performance

Model	Test R^2	Explained variance	Mean Euclidean distance	Median Euclidean distance	Mean cosine similarity	Median cosine similarity
Linear	−0.0349	−0.0305	1.6367	1.4404	0.0481	0.0936
Linear — shuffled control	−0.0714	−0.0655	1.6550	1.4282	0.0085	−0.0016
MLP	−0.2652	−0.2610	1.8432	1.6683	0.0339	0.0807
MLP — shuffled control	−0.2912	−0.2852	1.8445	1.6674	0.0148	0.0395

Higher is better for R^2, explained variance and cosine similarity; lower is better for Euclidean distance.

Figure 31 - Decoding the residual latent variable Z_1 of the one-dimensional hybrid model from the *readout feature weights*

4.5. Interpreting the residual coordinate Z1

Analysis of Z1 reveals a surprising result. When the $Z_1^{\mathrm{pre}} \times Z_1^{\mathrm{post}}$ space is discretised into quantile-defined classes and the empirical connection probability is calculated in each cell, it varies strongly with the latent value of the presynaptic neuron, but far less with that of the postsynaptic neuron. This asymmetry suggests that Z_1 mainly encodes an individual property of the presynaptic neuron.

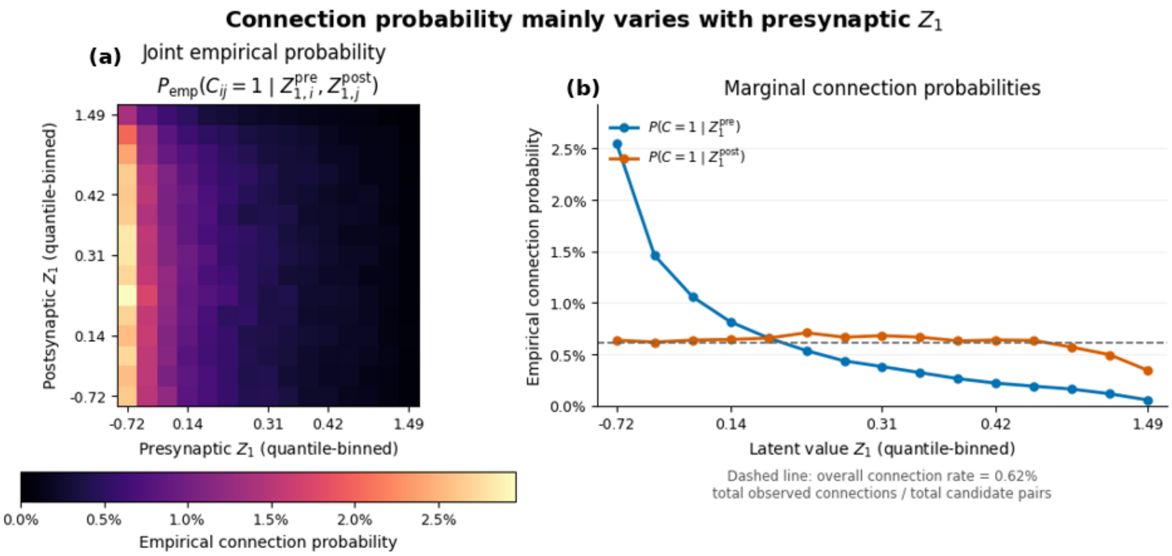


Figure 32 - Asymmetric dependence of connection probability on the latent coordinate Z_1. (a) Empirical connection probability as a function of presynaptic and postsynaptic values discretised by quantiles. (b) Marginal connection probabilities conditioned only on the presynaptic value (blue) or postsynaptic value (orange). The dotted line represents the overall observed connection rate, equal to 0.62%.

These results therefore suggest that Z_1 captures a factor inherent to each neuron that was not taken into account in our analysis and that allows its propensity to form connections to be estimated more accurately. This factor is not necessarily a single biological variable: it could also correspond to a technical bias.

To better understand the role of Z_1, the neurons were ordered according to their latent coordinate, then divided into equal-sized groups defined by quantiles. For each group, we calculated the mean in-degree and out-degree. The clearest trend concerns out-degree, which decreases sharply as Z_1^{pre} increases. This dimension therefore appears mainly to order presynaptic neurons according to their general propensity to establish connections.

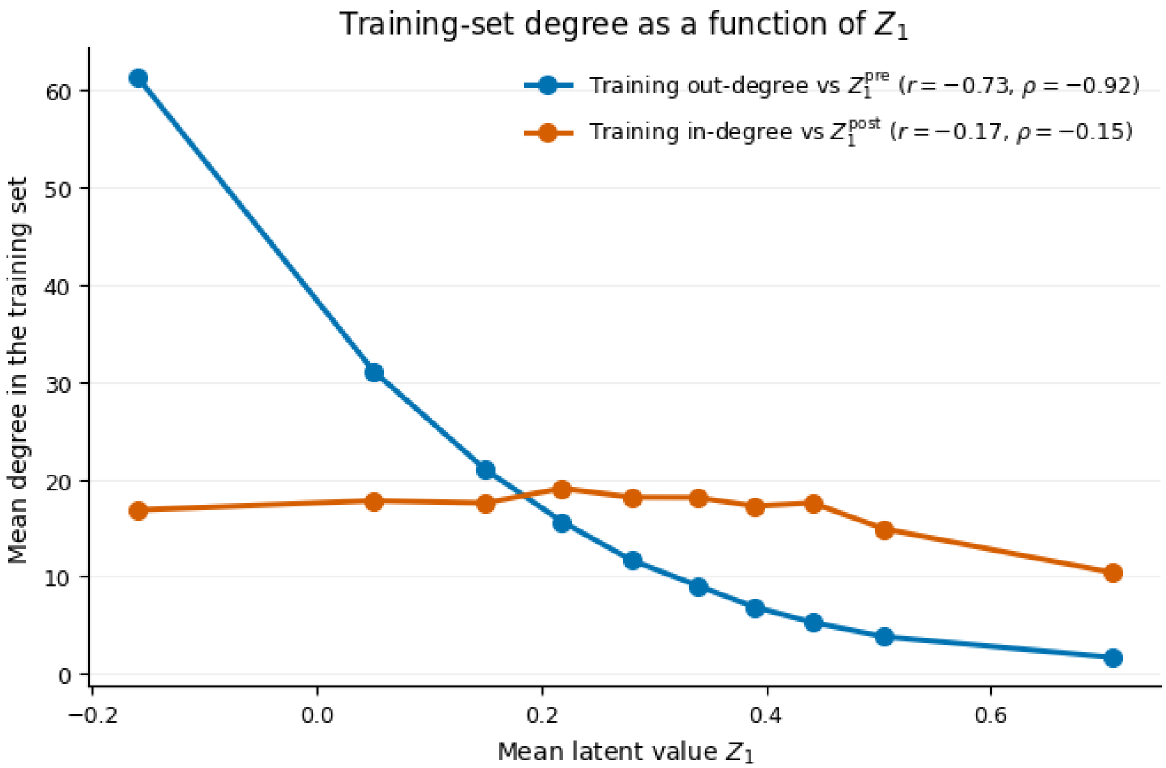


Figure 33 - Neuronal degrees as a function of the residual latent coordinate Z_1. Mean degrees calculated on the training set for groups of neurons defined by quantiles: out-degree as a function of Z_1^{pre} (blue) and in-degree as a function of Z_1^{post}(orange). r and ρ respectively denote the Pearson and Spearman correlation coefficients, calculated at the level of individual neurons before they were grouped by quantiles.

This observation is nevertheless insufficient to assign a biological meaning to Z_1. The same neurons appear in the training, validation and test sets, but their latent coordinates are learned only from the training connections. The model can therefore use a neuron's known connections to estimate its general propensity to connect, then apply this information to the 10% of its connections that remain unknown. This is therefore not data leakage, but a transductive evaluation on neurons that have already been observed.

It nevertheless remains difficult to determine whether Z_1 captures a biological, morphological or technical property that genuinely influences out-degree, or whether it simply represents the connection rate estimated from the training data. In the latter case, the model can improve the global AUC by distinguishing neurons that connect frequently from those that connect rarely, without necessarily identifying their postsynaptic partners more accurately. This limitation therefore mainly concerns the biological interpretation of latent space and its generalisation to new neurons.

4.6. Conditional AUC and control using out-degree

Two additional analyses strengthen the hypothesis that Z_1 mainly encodes the presynaptic neuron's general propensity to form connections. The global AUC compares the scores assigned to all connected and unconnected pairs. It can therefore be high if the model systematically assigns higher probabilities to pairs originating from presynaptic neurons with high out-degree, even without precisely identifying their postsynaptic partners.

To neutralise this effect, we calculated the AUC separately for each presynaptic neuron, then averaged the resulting values:

$$\mathcal{S}_i^{\text{pre}} = \{(y_{ij}, \hat{p}_{ij}) \mid j \in \mathcal{V},\ j \neq i\}.$$

$$\text{AUC}_i^{\text{pre}} = \text{AUC}(\mathcal{S}_i^{\text{pre}})$$

$$\text{AUC}_{\text{fixed-pre}} = \frac{1}{N_{\text{pre}}} \sum_{i=1}^{N_{\text{pre}}} \text{AUC}_i^{\text{pre}}$$

$\mathcal{S}_i^{\text{pre}}$ groups together all pairs associated with the same presynaptic neuron i. Since its general propensity to connect is common to all these pairs, it can no longer be used to distinguish between them. The model is therefore no longer evaluated on its ability to identify neurons that connect frequently, but on a more demanding question: predicting whom each of them connects to.

With this new metric, the hybrid model's performance falls sharply relative to its global AUC. In contrast, no comparable decrease appears when the postsynaptic neuron is fixed, nor with the reference geometric model based solely on spatial coordinates.

AUC summary

(a) Hybrid model

Evaluation	AUC	Median neuron AUC	95% CI lower	95% CI upper	Eligible neurons
Global test AUC	**0.9092**	—	—	—	—
Fixed presynaptic neuron	**0.8694**	0.8962	0.8644	0.8739	2,241
Fixed postsynaptic neuron	**0.9039**	0.9280	0.9005	0.9075	2,711

(b) Geometric model

Evaluation	AUC	Median neuron AUC	95% CI lower	95% CI upper	Eligible neurons
Global test AUC	**0.8604**	—	—	—	—
Fixed presynaptic neuron	**0.8631**	0.8911	0.8580	0.8678	2,241
Fixed postsynaptic neuron	**0.8503**	0.8731	0.8462	0.8545	2,711

Figure 34 - Comparison of the global and conditional AUCs of the hybrid and geometric models. Results obtained with the hybrid model (a) and the geometric model (b)

These results suggest that the gain of the hybrid model stems largely from its ability to distinguish presynaptic neurons that establish many connections from those that establish few. In contrast, once the presynaptic neuron is fixed, the latent representation improves the identification of its postsynaptic partners far less.

To test this interpretation, we finally constructed a model that takes as input the geometric positions of the neurons together with the out-degree of each presynaptic neuron, calculated only from the connections in the training set. The validation and test connections therefore remain entirely unknown when this variable is constructed. If this simple model achieves an AUC close to that of the hybrid model, it will indicate that a substantial proportion of the gain contributed by Z_1 can be reproduced by the neuron's general propensity to establish connections alone.

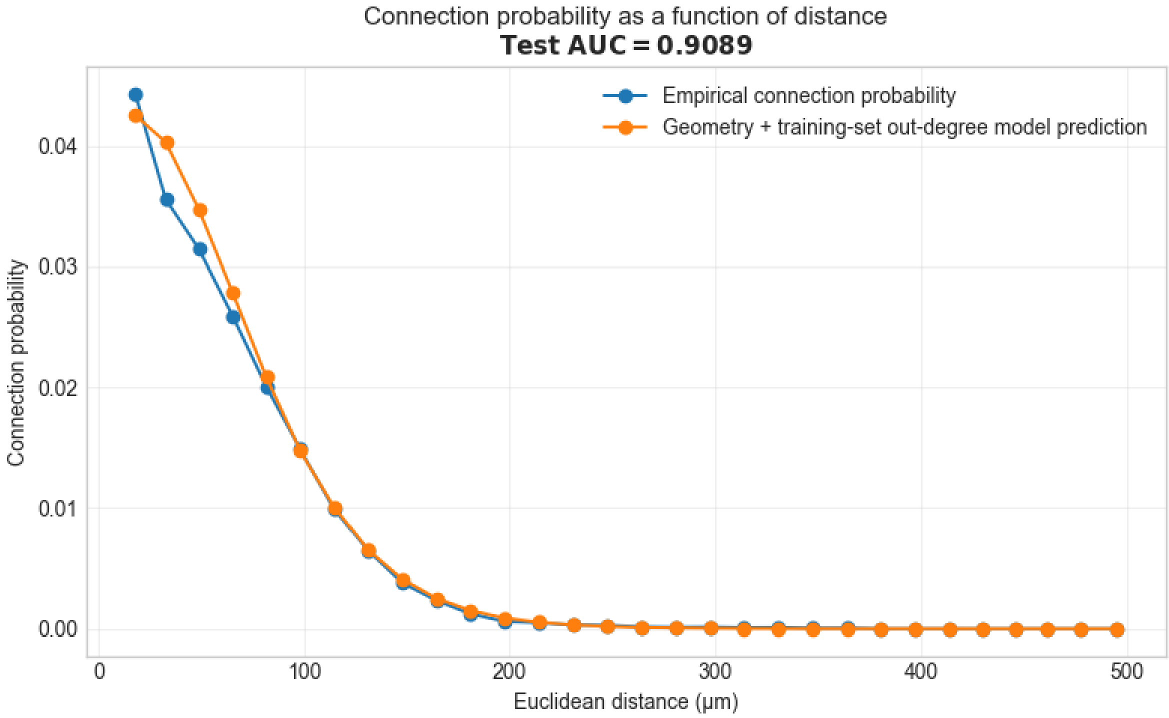


Figure 35 - Connection probability as a function of distance for the geometric model augmented with presynaptic out-degree. Empirical connection probability (blue) and mean connection probability predicted by the model (orange). The AUC shown is calculated over all pairs in the test set.

The model reaches a test AUC of 0.9089, almost identical to that of the hybrid model. This result confirms that most of the global predictive gain attributed to the residual latent variable can be reproduced using geometry together with the presynaptic out-degree estimated from the training set.

5. Critical analysis and future directions

5.1. Limits to the interpretation of latent representations

The preceding results highlight an important limitation in the interpretation of latent representations. Since out-degree is directly observable in the adjacency matrix, it is natural that a model trained to reconstruct that matrix should use it. It would be contradictory to ask the model to explain the connections as accurately as possible while forbidding it from exploiting one of their main statistical regularities. This behaviour therefore does not necessarily constitute a failure of the model, but a direct consequence of its learning objective.

Simply modifying the inputs would not, however, be sufficient to resolve this difficulty. As long as the objective remains the reconstruction of the adjacency matrix by minimising the BCE, the model is still encouraged to encode neurons' individual propensity to connect in whichever variables remain accessible to it.

5.2. Separating individual propensities from compatibility

One initial direction would be to separate explicitly three components of connectivity: neurons' individual propensities to form connections, the effect of their geometry and their residual compatibility. The connection score could, for example, be decomposed as follows:

$$\tilde{p}_{ij} = \sigma\left(\alpha_i^{\text{out}} + \beta_j^{\text{in}} + f_\phi(\mathbf{x}_i, \mathbf{x}_j) + g_\theta(\mathbf{Z}_i, \mathbf{Z}_j)\right)$$

The term α_i^{out} represents the neuron's general propensity to establish connections, while β_j^{in} describes the propensity of neuron j to receive them. The function f_ϕ models the effect of geometry explicitly from anatomical positions, while g_θ uses the latent representations to account for the connectivity structure that remains beyond these effects. The propensity terms would be estimated only from connections in the training set.

This separation between sender, receiver and interaction effects is notably present in the statistical models for dyadic data proposed by Hoff (2005) [14]. It is also consistent with the principle of degree-corrected network models, which distinguish degree heterogeneity from the relational structure being sought (Karrer and Newman, 2011) [15].

Explicitly introducing terms that describe individual propensities to emit or receive connections does not, however, guarantee that these effects will disappear from the latent representations. The model could still use the latter to increase or decrease the connection probability of certain neurons globally. To constrain the latent component to represent only compatibility between partners, one could therefore require its mean contribution to be zero for each neuron, on both the presynaptic and postsynaptic sides:

$$\left[\sum_j g_\theta\left(\mathbf{Z}_i, \mathbf{Z}_j\right) = 0, \qquad \sum_i g_\theta\left(\mathbf{Z}_i, \mathbf{Z}_j\right) = 0.\right]$$

The latent component could then no longer explain why one neuron forms more or fewer connections overall than another, but only differences in compatibility between partners. This constraint would not, however, guarantee the absence of all geometric information from the latent representations. Simultaneously eliminating the effect of individual propensities and the encoding of geometry in latent space represents a considerably more difficult challenge, one that would require substantial additional constraints.

5.3. Conditional ranking training

A second direction would be to retain the hybrid model but supplement the BCE with a conditional ranking loss. For the same presynaptic neuron (i), we consider a connected pair (i, j^+), such that $\left(y_{ij^+} = 1\right)$, and an unconnected pair (i, j^-), such that $\left(y_{ij^-} = 0\right)$. The scores assigned by the model before application of the sigmoid are then defined as follows:

$$s_{ij^+} = \mathrm{MLP}_\theta^{\mathrm{hybrid}}(\mathbf{x}_i, \mathbf{x}_{j^+}, \mathbf{Z}_i, \mathbf{Z}_{j^+}), \qquad s_{ij^-} = \mathrm{MLP}_\theta^{\mathrm{hybrid}}(\mathbf{x}_i, \mathbf{x}_{j^-}, \mathbf{Z}_i, \mathbf{Z}_{j^-})$$

We can then introduce a loss function that favours a higher score for the observed connection:

$$\mathcal{L}_{\mathrm{rank}}^{\mathrm{pre}} = -\log \sigma\left(s_{ij^+} - s_{ij^-}\right)$$

Since the presynaptic neuron is identical in both pairs, its general propensity to connect would not, on its own, make it possible to reduce this loss. The model would have to learn why j^+ is a more probable partner than j^-. This principle is close to the pairwise ranking methods developed notably by Rendle et al. (2009) [16]. A symmetric formulation could also be used by fixing the postsynaptic neuron.

The final loss function could then combine the two objectives:

$$L = \mathcal{L}_{\mathrm{BCE}} + \lambda_{\mathrm{pre}}\mathcal{L}_{\mathrm{rank}}^{\mathrm{pre}} + (\lambda_{\mathrm{post}}\mathcal{L}_{\mathrm{rank}}^{\mathrm{post}})$$

The BCE would continue to adjust the overall connection probability, while the conditional terms would encourage the learning of partner preferences. The weight assigned to each component would determine the trade-off between global prediction and the selection of synaptic partners.

5.4. Modelling reconstruction uncertainty

One final direction concerns the joint use of the connectome's different reconstruction levels. Until now, we have mainly sought to limit the influence of proofreading by working on subsets with a relatively homogeneous level of reconstruction. A more general approach would instead be to incorporate this uncertainty directly into the model.

An idea proposed by Prof. Sanzeni is to distinguish the probability that a connection exists biologically from the probability that it is actually observed in the connectome. The probability of observing a connection between the neurons could thus be written as:

$$\tilde{P}_{\mathrm{obs}}(i,j) = \tilde{P}_{\mathrm{conn}}(i,j)\left(1 - \tilde{P}_{\mathrm{err}}(i,j)\right)$$

The first term $\tilde{P}_{\mathrm{conn}}$ represents the connection probability estimated by the model. It could notably correspond to the hybrid model developed above, although this formulation remains compatible with the other models studied in this work. The second term, $\tilde{P}_{\mathrm{err}}(i,j)$, represents the probability that an existing connection is not observed because of an incomplete reconstruction.

This error probability would itself be estimated from the distance between the neurons and the available information about their proofreading level:

$$\tilde{P}_{error}(i,j) \;=\; \sigma\left(MLP_{error}\left(\left\|\boldsymbol{x_i} \;-\boldsymbol{x_j}\right\|, \boldsymbol{A}_i, \boldsymbol{D_j}\right)\right)$$

where $\boldsymbol{x_i}$ and $\boldsymbol{x_j}$ denote the three-dimensional positions of the neurons, and $\mathbf{A}_i \in \{0,1\}^4$ one-hot encodes the reconstruction level of the presynaptic axon and $\boldsymbol{D_j} \in \{0,1\}^3$ that of the postsynaptic dendrite. The dependence on distance reflects the idea that automatic segmentation errors become more likely as the distance between neurons increases.

When the presynaptic axon is fully reconstructed, we impose in particular that:

$$\tilde{P}_{error}(i,j) = 0$$

This model would thus make it possible to exploit the entire connectome jointly, in particular by using CINECA's computing resources. A few exploratory trials carried out at the centre of the connectome, where fully reconstructed and unreconstructed neurons coexist, using the hybrid model as the connection-probability estimator, have already produced encouraging results.

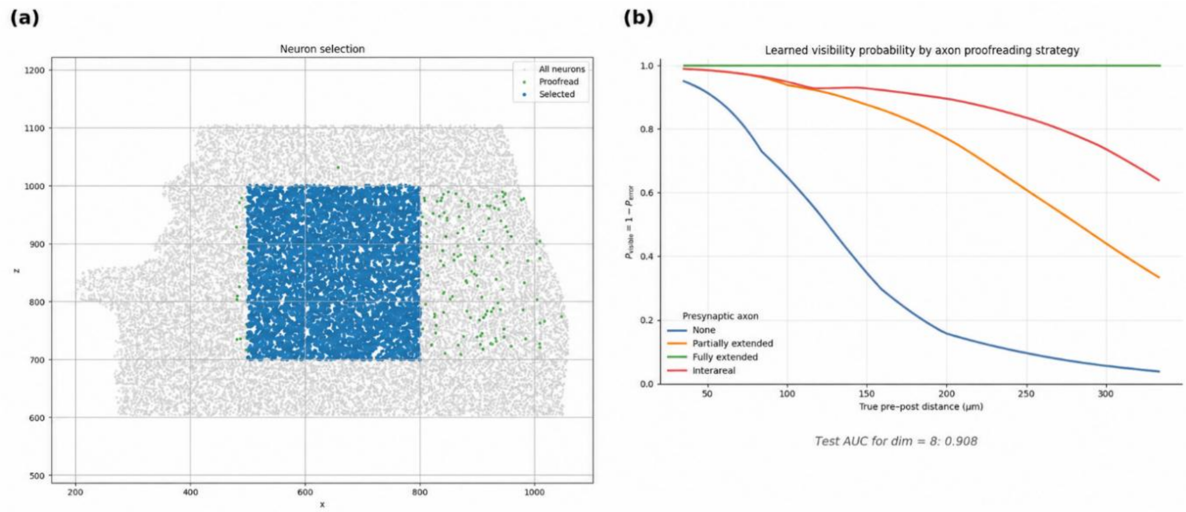


Figure 36 - Modelling reconstruction uncertainty. Selection of the analysis region at the centre of the connectome. Grey points represent all neurons, green points the proofread neurons and blue points the neurons retained for analysis (a) Estimated visibility probability as a function of pre-post distance for the different proofreading strategies applied to the presynaptic axon (b)

The 8-dimensional hybrid model incorporating reconstruction uncertainty achieves a test AUC of 0.908, comparable to that obtained without this correction. The learned visibility probabilities also exhibit consistent behaviour: they decrease with distance, except at the highest reconstruction level, for which the imposed constraint maintains visibility at 1. The expected hierarchy between the different reconstruction levels is also respected: visibility is lowest for automatic segmentation, intermediate for manually partially reconstructed axons and highest for fully reconstructed axons. An identifiability problem nevertheless remains: without additional constraints, some of the connectivity information could be absorbed by the uncertainty estimator, and vice versa.

6. Conclusion

In conclusion, this new method to uncover the connectome's connectivity rules, based on constructing and learning a latent space from the adjacency matrix, produces promising results. In particular, it recovers the predominant role of spatial proximity in the organisation of the network. In contrast, the other biological and functional variables studied, such as orientation preference and the *readout features*, produce a weaker signal.

The flexibility of this approach, which allows it to capture complex information without imposing rules a priori, nevertheless also constitutes its main weakness. The model appears particularly sensitive to statistical biases, whether they arise from inhomogeneities in neuronal reconstruction or from learning a simple individual propensity to establish connections rather than the biological mechanisms that genuinely determine them. This latter hypothesis remains to be confirmed, but several results appear to converge in this direction: the distribution of the predicted probabilities seems to depend mainly on the presynaptic neuron, the AUC decreases markedly when it is computed separately for each presynaptic neuron, and a model combining geometry and out-degree alone achieves performance comparable to that of the hybrid and latent models.

At this stage, however, it remains impossible to determine whether this propensity to connect corresponds to a simple statistical bias learned by the model or to a genuine biological property absent from the currently available variables. Despite their diversity, these variables are probably not exhaustive. One could readily imagine the existence of subpopulations of particularly well-connected neurons that might be described as "super-neurons" playing a specific role in the mouse visual cortex.

This approach therefore constitutes a promising tool for bringing to light still-unknown properties of connectome organisation. Its main challenge now lies in refining the models and their learning objectives: the aim is to prevent dominant signals from masking more subtle information, without eliminating those signals or losing the relevant biological information they contain.

Finally, our analysis focused mainly on connectivity rules between pairs of neurons. Other work nevertheless suggests that network structure cannot be reduced to these pairwise interactions. Song et al. (2005) [17] notably show that certain motifs involving three neurons are overrepresented: when neuron A is connected to B and B to C, the probability of also observing a connection between A and C becomes higher than expected in a random network. Studying such higher-order dependencies therefore constitutes another natural direction in which to extend this work and illustrates how much of the fine organisation of cortical connectivity still remains to be explored.

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Abstract

This work investigates the statistical rules that structure connectivity in the mouse visual cortex using the MICrONS connectome. The objective is to determine whether a latent space learned solely from the adjacency matrix can recover known anatomical or functional properties and reveal previously uncharacterised connectivity factors. After reproducing a geometric model linking intersomatic distance to connection probability, we adapt an embedding method originally developed to identify symmetries in connectomes. Each neuron is assigned a latent representation, jointly optimised with a multilayer perceptron that reconstructs binary connections.

The experiments show that the latent space spontaneously recovers the geometry of the cortical volume: spatial proximity is the main determinant of connectivity, whereas orientation preference and functional readout variables generate a weaker signal. A hybrid model receiving anatomical coordinates explicitly reaches almost the same performance as the latent model with a single residual dimension. Analysis of this dimension, however, reveals a strong dependence on each presynaptic neuron's individual propensity to establish connections. The conditional AUC, computed separately for each presynaptic neuron, drops substantially, while a model combining geometry with presynaptic out-degree reproduces most of the observed improvement.

These results support the use of latent representations as an exploratory tool while highlighting their sensitivity to reconstruction biases and simple statistical regularities. Proposed extensions include separating individual propensity, geometry and partner compatibility, incorporating reconstruction uncertainty and investigating higher-order connectivity motifs.

Keywords: MICrONS connectome; cortical connectivity; latent representations; adjacency matrix; machine learning; reconstruction bias.